HARMONIC PATHS AND HONEYCOMBS

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Abstract. We construct harmonic paths of non-expansive norms on the identity representation of the A_2 quiver from arbitrary triples of diagonalizable boundary norms over the real-exponent Novikov field. We use symmetric central charges and a fixed radial area form on the plane. In every finite rank, the path has a finite apartment presentation and a finite honeycomb support with actual semistable edge labels and compatible filtrations at its vertices. It is unique among paths with such presentations whose one-sided norm motions agree to first order at each join. The construction minimizes a renormalized convex action. Its algebraic regularity follows from fixed orthogonal subspaces at gaps in the spectral velocities, second-order scalar approximations, and descent of finite norm data from the Hahn field. Repeated boundary spectra and collapsed edges are included. The result gives existence in the honeycomb specialization of the harmonic-path conjecture of Haiden–Katzarkov–Kontsevich–Pandit, with uniqueness under an explicit matching condition absent from the printed edgewise definition.

1. Introduction

A triple of non-Archimedean norms contains more information than the three relative spectral lists. The harmonic-path problem of Haiden–Katzarkov–Kontsevich–Pandit asks for a geometric reconstruction that retains this additional information: a path of non-expansive norms whose spectral velocities trace a labelled network in the plane [HKKP26, §§7.2–7.3]. We construct such paths for the identity representation of the A_2 quiver, for every full boundary triple and in every finite rank. The principal point is finite algebraic regularity of a variational minimum in a completed norm space. A separate converse identifies that minimum among paths satisfying an explicit first-order matching condition.

1.1. **The boundary problem.** Let K be the real-exponent Novikov field over the trivially valued field $\mathbb{C}$. Thus its elements are formal sums

$$f = \sum_{\lambda \in \mathbb{R}} a_\lambda T^\lambda$$

whose support is bounded below and has finite intersection with $(-\infty, b]$ for every $b \in \mathbb{R}$. For $f \neq 0$ set $\nu(f) = \min\{\lambda : a_\lambda \neq 0\}$ and $|f| = e^{-\nu(f)}$. We write $\mathcal{O} = \{|f| \leq 1\}$ and $\mathfrak{m} = \{|f| < 1\}$; the residue field is $\mathcal{O}/\mathfrak{m} = \mathbb{C}$. This field is complete but not spherically complete. Its spherically complete Hahn extension will be used in the proof; the path constructed by the theorem is defined over K.

Fix $V = K^n$ and consider the quiver with one arrow from its vertex C to its vertex A. We use the representations

$$A = (0 \longrightarrow K), \qquad B = (K \xrightarrow{\mathrm{id}} K), \qquad C = (K \longrightarrow 0), \qquad E = B^n, \tag{1}$$

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and the central charges

$$Z(A) = e^{\pi i/6}, \qquad Z(B) = i, \qquad Z(C) = e^{5\pi i/6}. \tag{2}$$

A norm h on V is *diagonalizable*, or *split*, if there are a basis $e_1, \dots, e_n$ and real weights $w_1, \dots, w_n$ such that

$$h\Big(\sum_i b_i e_i\Big) = \max_i e^{w_i}|b_i|. \tag{3}$$

The norms diagonalized by a fixed basis form an apartment. We retain their scales; norms are not taken modulo multiplication by a positive scalar. Two split norms admit a common orthogonal basis. Their d_2 distance is the Euclidean norm of the differences of their logarithmic weights in such a basis. In particular, this is the unnormalized metric, including the determinant direction.

A norm on E is a pair $c = (c_A, c_C)$. The identity arrow is non-expansive exactly when

$$c_A \le c_C \tag{4}$$

pointwise on V. For a split norm h, put $\overline{V}_h = \{h \le 1\}/\{h < 1\}$. A non-expansive pair induces an arrow $\overline{V}_{c_C} \to \overline{V}_{c_A}$ over $\mathbb{C}$. The harmonicity condition for a norm in [HKKP26, Definition 7.3] is semistability of this reduced representation. Every representation of this one-arrow quiver over $\mathbb{C}$ is a direct sum of A, B and C. Their phases are distinct, so the nonzero semistable objects are precisely A^m, B^m and C^m.

The boundary datum in this paper is an arbitrary triple of split norms

$$(h_B, h_A^+, h_C^+) \quad \text{on the fixed space } V. \tag{5}$$

It determines the trivial negative-end filtration of E and the positive-end filtration $0 \subset A^n \subset E$, with quotient C^n. Indeed, a non-expansive norm on B^n is harmonic precisely when its reduced arrow is invertible. In a common orthogonal basis for its two norms, this says that every relative weight is zero, hence that the two norms agree. Thus h_B is exactly a harmonic norm on the negative-end object, while h_A^+ and h_C^+ are independent harmonic norms on the two positive-end factors. No initial comparison between these three norms is required.

We fix the area form and its transverse coordinate

$$\omega = \frac{\mathrm{d}x \wedge \mathrm{d}t}{1 + x^2 + t^2}, \qquad \lambda(x,t) = \frac{1}{\sqrt{1+t^2}} \arctan \frac{x}{\sqrt{1+t^2}}. \tag{6}$$

Let

$$\begin{aligned} a &= \sqrt{3}, \qquad t = \sinh u, \qquad R(u) = \cosh u, \\ \beta(u) &= \arctan(a \tanh u), \qquad F(u) = \int_0^u \beta(s)\,\mathrm{d}s. \end{aligned} \tag{7}$$

We require the following limits in d_2:

$$\begin{aligned} c_A(u), c_C(u) &\longrightarrow h_B \quad (u \longrightarrow -\infty), \\ e^{F(u)} c_A(u) &\longrightarrow h_A^+ \quad (u \longrightarrow +\infty), \\ e^{-F(u)} c_C(u) &\longrightarrow h_C^+ \quad (u \longrightarrow +\infty). \end{aligned} \tag{8}$$

These are the HKKP normalizations. In fact, for $A_\mu(t) = \int_0^t \lambda(\mu s, s)\,\mathrm{d}s$, one has $A_a(\sinh u) = F(u)$ and $A_{-a} = -A_a$. The conjugate-charge direction convention in [HKKP26, Definition 7.8] gives slopes $-a, 0, a$ for the labels A, B, C, respectively.

1.2. **First-order matching and the main theorem.** We specify the comparison class in the uniqueness statement. This is necessary because edge directions and abstract edge labels do not by themselves identify the embeddings of the velocity subspaces at a join.

Definition 1.1. A *finite regular harmonic path* is a continuous ordered pair of split norms on the whole u-line with the following properties. There is a finite subdivision into intervals on each of which both norms lie in fixed apartments, with weight functions smooth up to the finite endpoints. After subdivision at the intersections of the support branches, the weights are grouped by their derivative functions $p_g(u) \in (-\pi/2, \pi/2)$, and

$$R(u)\tan p_g(u) = \alpha_g \sinh u + b_g, \qquad \alpha_g \in \{-a, 0, a\}, \quad b_g \in \mathbb{R}. \tag{9}$$

Within each group the relative weights and reduced arrow are constant. The actual reduced object is semistable, of type A^{m_g}, B^{m_g} or C^{m_g}, according as $\alpha_g = -a, 0, a$. The closure of these labelled branches is the support of the path.

The path satisfies *first-order matching* if, at every finite subdivision time u_0 and for $v = A, C$,

$$d_2\big(c_v(u_0), \mathrm{mid}(c_v(u_0 - h), c_v(u_0 + h))\big) = o(h) \qquad (h \downarrow 0). \tag{10}$$

Here mid is the midpoint of the canonical norm geodesic. This condition is independent of bases and of artificial subdivisions.

In a single apartment, (10) is equality of the two one-sided first derivatives. More generally, it requires the one-sided norm motions to fit to first order in the metric space of norms. The groups in Definition 1.1 retain their actual reduced arrows, not merely their dimensions. In particular, a group with both source and target is of type B^{m_g} only when its reduced arrow is an isomorphism.

Theorem 1.2. *Let $n \geq 1$. For every triple (5) of split norms over K, there is a unique finite regular harmonic path satisfying first-order matching and the normalized ends (8). It has the following properties.*

1. *Its support is a finite balanced weighted honeycomb with incoming B-multiplicity n and outgoing A- and C-multiplicities n each. It has at most n^2 essential geometric vertices.*
2. *At every support vertex there is an actual reduced quiver representation with two increasing-phase filtrations. Their successive quotients are the incident incoming and outgoing edge objects. These identifications agree along the edges and remove artificial subdivisions.*
3. *Each tail lies in fixed K-apartments. The two norms agree for all sufficiently negative times. All three normalized limits have d_2 error $O(|t|^{-1})$.*
4. *The path is equivariant under simultaneous $\mathrm{GL}_n(K)$ change of basis of the three boundary norms.*

The path has analytic one-sided weights and satisfies the stronger $O(h^2)$ estimate in (10).

The multiplicities in the theorem specify the degree of the honeycomb; they do not prescribe the valence of its vertices. In particular, crossing vertices can be four-valent. The n^2 count is the standard finite honeycomb bound from the finite dual subdivision in Knutson–Tao theory [KT99, Lemmas 4 and 6–8 and Theorem 1]. The analytic content is that the variational construction has finite support to which this bound applies, and then a finite apartment description over the original Novikov field. The theorem includes repeated boundary spectra,

coincident edges and collapsed bounded edges. Its estimates need not be uniform over all boundary triples.

The uniqueness is equality of the actual norms at every time. It is stronger than equality of support or S-equivalence within the stated class. The theorem does not claim that every honeycomb with the same boundary spacings gives the same path. Nor does the inclusion of degenerate inputs by itself assert continuous dependence on arbitrary boundary data. An explicit rank-three family below has a bounded face that collapses, with a uniform Lipschitz estimate for its full norm paths.

1.3. **Relation to the harmonic-path conjecture.** HKKP conjecture a correspondence between harmonic paths and pairs of ascending-phase filtrations with harmonic norms on their quotients [HKKP26, Conjecture 7.10]. Their honeycomb example identifies the boundary data with triples of diagonalizable norms [HKKP26, §7.3]. Theorem 1.2 gives existence for every such triple in the present field and area form. Its uniqueness statement uses the finite regular class and the explicit condition (10).

The displayed conditions of [HKKP26, Definition 7.8] do not explicitly impose this matching. They permit coordinate velocity subspaces to switch at finitely many times while the support and abstract edge labels remain unchanged. We give two rank-three examples with the same full normalized ends and the same connected balanced support that differ by such switches. They fail (10). This explains why we do not state Theorem 1.2 as an unqualified proof of Conjecture 7.10. The construction also supplies the local residue-category vertex data, but we do not prove that abstract vertex filtrations alone imply (10), or identify every globally framed categorical network with the norm paths in the theorem.

Haiden–Katzarkov–Simpson construct the category of spectral networks with vertex filtrations and edge identifications [HKS24, §4.4]. In their constant-coefficient, product-type setting, the uniqueness theorem gives the same support and S-equivalent reductions for networks representing the same object after Novikov-field extension [HKS24, Theorem 4.15]. Their regular-hexagon theorem supplies spectral network representatives in a specified A_2 setting [HKS24, Theorem 6.1]. These are different assertions from reconstruction of the full framed norm path from arbitrary boundary norms. The family-Floer comparison discussed in [HKKP26, Remark 7.9] is not used in our existence or uniqueness proof.

The relation between honeycombs and feasible spectral triples is classical [KT99]; Moitra–Postnikov–Woodruff give an axiomatic treatment of that relation [MPW26]. Appleby–Whitehead construct real-valued Littlewood–Richardson fillings from matrix pairs over valuation rings and hives from pairs of full lattices [AW13, AW18]. These constructions also retain information beyond pairwise spectra. Our claim concerns the harmonic norm path for the specified area form, rather than a first construction of numerical honeycombs from valued linear algebra. Ben-Bassat's split attainment theorem [BB26, Theorem 1.1] concerns the stationary harmonic-norm potential of HKKP. Split attainment at each individual time does not supply a coherent path or finite apartment structure; those require the regularity argument below.

1.4. **The variational argument.** HKKP suggest a variational approach based on a convex function of the velocity [HKKP26, p. 85]. For (6), its scalar kinetic term in the u coordinate is

$$\ell(v) = -\log\cos v, \qquad |v| < \pi/2.$$

In a product apartment the local action is

$$\int_I R(u)\left\{\frac{1}{2}\sum_{v=A,C}\sum_i \ell(w'_{vi}(u)) + \frac{a}{2}\big(d(c_C(u)) - d(c_A(u))\big)\right\}\mathrm{d}u, \tag{11}$$

where $d(h)$ is the logarithmic determinant norm in a common fixed volume normalization. The positive-end reference velocities are $-\beta$ and β. Subtracting their supporting affine terms gives a nonnegative renormalized action on the whole line. A weighted strict convexity estimate makes every minimizing sequence uniformly Cauchy. The limit is first constructed in the complete space of all norms.

The main difficulty is to prove that this limit is a finite algebraic path. Short ordered replacements give quadratic approximation by endpoint geodesics and locally Lipschitz spectral velocities. Where groups of velocities are separated, equality in the relevant Ky Fan inequalities forces fixed orthogonal subspaces. At almost every time, scalar normalization on these subspaces gives second-order norm approximations. An endpoint repair with action error $o(h^3)$ permits variations by actual residue subrepresentations. The resulting equations force semistability and the three honeycomb directions.

The direction laws imply that each opposite-type pair of spectral positions has at most one contact interval. There are consequently only finitely many support events. On every interval between them the fixed subspaces have scalar norm motion, giving a single apartment. A finite valued-field transfer preserves the full endpoint norms, the reduced arrows and the action, and identifies the resulting K-path with the original minimum. Finally, opposite momentum directions at a matched join give nonnegative Busemann boundary terms. This proves the converse across all joins and yields the equality uniqueness in Theorem 1.2.

2. Norm geometry and finite realization

We use the complete space of all norms to construct a minimum and return to split norms only after establishing finite regularity. This section records the spectral convexity, finite realization and projection results needed for that argument.

2.1. **Relative spectra and the completion.** For a finite-dimensional space W over a complete non-Archimedean field, write $\mathcal{N}_{\mathrm{diag}}(W)$ for the split norms and $\mathcal{N}(W)$ for all norms. The Goldman–Iwahori distance is

$$d_\infty(h,k) = \sup_{z\neq 0}\left|\log\frac{k(z)}{h(z)}\right|.$$

If h, k are split, choose a common orthogonal basis and arrange the logarithms $\log(k(e_i)/h(e_i))$ in decreasing order:

$$\sigma(h,k) = (\sigma_1(h,k),\dots,\sigma_n(h,k)).$$

Put $S_j(h,k) = \sum_{i=1}^{j}\sigma_i(h,k)$ and $S_0 = 0$. Equivalently, S_j is the logarithm of the operator norm of the identity on the jth exterior power, with its induced determinant norms. This characterization extends S_j continuously to all norms. Indeed, if $d_\infty(h,h') \le \varepsilon$ and $d_\infty(k,k') \le \eta$, comparison of exterior norms gives

$$|S_j(h,k) - S_j(h',k')| \le j(\varepsilon+\eta). \tag{12}$$

We define the spectral coordinates on all norms by $\sigma_j = S_j - S_{j-1}$.

Fix a nonzero volume vector $\xi \in \bigwedge^n W$ and let $d(h)$ be the logarithm of its determinant norm. For a basis matrix P relative to a fixed frame with volume ξ, and weights w_i, this is

$$d(h) = \sum_i w_i + \nu(\det P). \tag{13}$$

Thus

$$S_n(h,k) = d(k) - d(h), \qquad d_2(h,k)^2 = \sum_i \sigma_i(h,k)^2, \qquad d_\infty \le d_2 \le \sqrt{n}\, d_\infty. \tag{14}$$

Changing ξ adds the same constant to all determinant logarithms. Only differences with a common volume convention will occur.

The canonical geodesic between two split norms interpolates their logarithmic weights in a common orthogonal basis. Split norms are dense in $\mathcal{N}(W)$; the geodesics and the spectral metric extend to the completion. The space $(\mathcal{N}(W), d_2)$ is complete and CAT(0), and these are its unique geodesics [BE21, Theorem 1.19 and Corollary 3.3]. We always mean these geodesics, even when discussing a different spectral metric. Determinant logarithms are affine along them. The existence of a common basis for two split norms, as well as the restriction and quotient properties of split norms, is given in [BE21, Lemma 1.13 and Proposition 1.14].

For $W = V$, use the product metric $d((h_A, h_C), (k_A, k_C))^2 = d_2(h_A, k_A)^2 + d_2(h_C, k_C)^2$ when applying CAT(0) arguments. Suprema in time will be taken with the equivalent maximum of the two vertexwise distances. Let

$$\mathcal{D} = \{(h_A, h_C) \in \mathcal{N}(V)^2 : h_A \le h_C\}.$$

Lemma 2.1. *Let h_θ, k_θ be canonical geodesics. Then*

$$\sigma(h_\theta, k_\theta) \prec (1-\theta)\sigma(h_0, k_0) + \theta\sigma(h_1, k_1), \qquad 0 \le \theta \le 1, \tag{15}$$

where $\prec$ denotes majorization with equality of total sums. Consequently, for every lower-semicontinuous convex function $\phi : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$,

$$(h,k) \longmapsto \sum_i \phi(\sigma_i(h,k)) \tag{16}$$

is jointly geodesically convex and lower semicontinuous. The set $\mathcal{D}$ is closed and geodesically convex. Moreover,

$$\sigma(h,l) \prec \sigma(h,k) + \sigma(k,l). \tag{17}$$

Proof. First suppose the endpoints are split. Fix one basis for the geodesic h_θ and another for k_θ. The logarithm of the norm of an exterior power of the identity is the maximum of finitely many affine functions of the two sets of weights, indexed by its nonzero minors. Thus every $S_j(h_\theta, k_\theta)$ is convex; this is the minor proof of [HKKP26, Lemma 6.16]. For $j = n$, the determinant identity gives affineness. The weighted sum on the right of (15) is decreasing, so these are exactly the majorization inequalities. The convex-hull characterization of majorization and scalar Jensen's inequality give (16). Neither monotonicity nor evenness of ϕ is needed. If ϕ is finite only on an interval, finite endpoint costs and majorization keep the intermediate spectra in that interval; the extended-real inequality follows as well.

Approximate the endpoints by split norms. Continuity of geodesics and (12) extend majorization to all norms. The convexity conclusion follows from the same majorization inequality, and lower semicontinuity follows from that of ϕ. The order condition is equivalent to $\sigma_1(h_C, h_A) \le 0$, so its sublevel set is closed and convex. Finally, submultiplicativity of exterior operator norms gives $S_j(h,l) \le S_j(h,k) + S_j(k,l)$; equality for $j = n$ follows from (14). This proves (17). □

We also use the following quantitative CAT(0) inequality. For arbitrary x, x', y, y' and their respective midpoints m_x, m_y,

$$d(m_x, m_y)^2 \le \frac{d(x,y)^2 + d(x',y')^2}{2} - \frac{\big(d(x,x') - d(y,y')\big)^2}{4}. \tag{18}$$

It is a form of the energy convexity inequality of Korevaar–Schoen [KS97, Proposition 1.3, equation (1.3.2)]. To see the coefficient, write $b = d(x,x')$, $c = d(y,y')$. Two applications of the CAT(0) midpoint inequality bound $4d(m_x, m_y)^2$ by

$$d(x,y)^2 + d(x,y')^2 + d(x',y)^2 + d(x',y')^2 - b^2 - c^2.$$

The CAT(0) quadrilateral inequality bounds the two cross terms by $d(x,y)^2 + d(x',y')^2 + 2bc$, giving (18). No local compactness is required.

2.2. **Scalar extension and finite realization.** The field K is algebraically closed [FOOO10, Appendix A, Lemma A.1]. Let

$$L = \mathbb{C}((T^{\mathbb{R}}))$$

now denote the Hahn field, whose supports are arbitrary well-ordered subsets of $\mathbb{R}$. It is algebraically closed and spherically complete [Poo93, Lemma 4, Theorem 1 and Corollary 4]. The natural inclusion $K \subset L$ preserves the value group and residue field. Model completeness of algebraically closed valued fields makes this inclusion elementary. In particular, a finite existential condition over K holding in L holds in K; see [Yin10, Definitions 2.1–2.2 and Theorem 3.10] for the valued-field language and quantifier-elimination statement.

For $h \in \mathcal{N}(W)$, its canonical extension to W_L is

$$h_L(z) = \inf_{z=\sum_i b_i z_i} \max_i |b_i| h(z_i), \qquad b_i \in L, \quad z_i \in W. \tag{19}$$

This is a norm, restricts to h, is isometric for d_∞, and preserves orthogonal bases and their weights [BE21, Definition 1.24 and Proposition 1.25]. Every finite-dimensional L-norm is split [BE21, Lemma 1.12]. Approximating a pair of K-norms by split norms and using a common basis for the approximants shows, by (12), that canonical extension preserves their entire relative spectrum. The same argument applies to an invertible K-linear arrow by pullback. Consequently extension is an isometry for d_2 and preserves canonical geodesics and order.

Lemma 2.2 (Finite realization)**.** *Let a finite directed graph have an n-dimensional K-space V_v at each vertex and a fixed K-linear isomorphism T_e on each edge. Give each V_v an arbitrary norm h_v, and specify a subset of vertices at which the norms are split. There are split K-norms k_v which agree exactly with the prescribed norms at those vertices and preserve the entire logarithmic singular-value list of every T_e.*

Proof. Extend the diagram canonically to L, preserving every spectrum. Choose orthonormal L-basis matrices P_v, using fixed orthonormal K-basis matrices at the specified vertices. Orthonormal bases exist because the norms are split over L and both value groups are $\mathbb{R}$. Introduce matrices Q_v with equations $P_v Q_v = Q_v P_v = I$. The arrow in these bases is $M_e = Q_{t(e)} T_e P_{s(e)}$.

For each $1 \le j \le n$ set $c_{e,j} = -\sum_{i=1}^{j} \sigma_i(T_e)$. Preserving the first j singular values in sum means precisely that every j-minor of M_e has valuation at least $c_{e,j}$ and one such minor has valuation exactly $c_{e,j}$. These are finite valued-field conditions with the parameters $T^{c_{e,j}} \in K$: every normalized minor belongs to $\mathcal{O}$, and at least one is a unit. The boundary matrices and their inverses are fixed, not merely the lengths of their columns.

The chosen L-matrices solve this finite formula. Elementarity gives K-matrices solving the same formula. Declare their columns orthonormal. The resulting norms have the required full spectra by the exterior-minor formula, and the fixed boundary matrices give exactly the prescribed boundary norms. □

Only finitely many real exponents are parameters in this argument. It does not quantify over norm functions or over a continuum of time values. Nor does it assert proximity between the newly realized norm and the original norm at an unspecified vertex. These distinctions will matter when passing to a limit of polygons.

2.3. **Projections associated with a decomposition.** Fix a direct sum $V = V_1 \oplus \cdots \oplus V_m$. Let $\mathcal{F}$ be the set of norms for which this decomposition is orthogonal:

$$h(z_1 + \cdots + z_m) = \max_i h(z_i), \qquad z_i \in V_i. \tag{20}$$

This permits arbitrary norms on the summands and does not require each summand norm to be split. If the summands are lines, $\mathcal{F}$ is the corresponding complete apartment.

Lemma 2.3. *The nearest-point projection $P : \mathcal{N}(V) \to \mathcal{F}$ preserves norm order and determinant logarithms, sends split norms to split norms, and satisfies*

$$\sum_i \phi\big(\sigma_i(Ph, Pk)\big) \le \sum_i \phi\big(\sigma_i(h, k)\big) \tag{21}$$

for every lower-semicontinuous convex scalar function ϕ.

Proof. Let G be generated by the signs s_i which act as -1 on V_i and as 1 on the other summands. Since $|2| = 1$, a G-invariant norm makes the coordinate projection $(I - s_i)/2$ nonexpansive. Together with the ultrametric triangle inequality this proves (20). The converse is immediate. Thus $\mathcal{F}$ is the common fixed set of a finite group of isometries; it is nonempty, closed and convex.

The unique barycenter of the orbit $\{g_*h : g \in G\}$ belongs to $\mathcal{F}$ by invariance. On $\mathcal{F}$ its squared-distance objective is $|G| d_2(z, h)^2$, so it is precisely Ph. A finite squared-distance barycenter of points in a closed convex set lies in that set. Indeed, projection to the set decreases every squared distance by at least the squared distance of the barycenter to its projection. This would strictly decrease the objective unless the barycenter already belonged to the set.

If $h_A \le h_C$, the orbit pairs belong to the closed convex order set $\mathcal{D}$. Their product barycenter is (Ph_A, Ph_C), so order is preserved. The determinant level set through h is closed and convex and contains its orbit, because $|\det g| = 1$. Hence $d(Ph) = d(h)$. By Lemma 2.1, every finite sublevel set of (16) is closed and convex. The orbit pairs (g_*h, g_*k) have the same relative spectra and belong to this set. Taking their barycenter proves (21). The case of infinite right-hand side is automatic.

Finally suppose h is split. Its orbit norms are split. Apply Lemma 2.2 to the unknown norm Ph, the fixed orbit norms, and the identity arrows joining it to each orbit norm. There is a split norm with the same distances to all orbit norms and therefore the same barycenter objective. Uniqueness identifies it with Ph. □

The contraction concerns the full convex spectral cost, rather than only d_2. We shall also need its quantitative part. Extend $\ell(v) = -\log\cos v$ by $+\infty$ for $|v| \ge \pi/2$. For $\Delta, W > 0$ set

$$\mathcal{K}_{\Delta,W}(h, k) = \frac{W}{2} \sum_i \ell\big(\sigma_i(h, k)/\Delta\big).$$

Lemma 2.4. *For a finite-cost pair h, k, the projection of Lemma 2.3 satisfies*

$$\mathcal{K}_{\Delta,W}(h,k) - \mathcal{K}_{\Delta,W}(Ph,Pk) \geq \frac{W}{8\Delta^2}\big(d_2(h,\mathcal{F}) - d_2(k,\mathcal{F})\big)^2. \tag{22}$$

Proof. For any nearest-point projection onto a closed convex subset of a complete CAT(0) space, the projection is constant along the segment from a point to its projection. To check this, let $q = Px$ and let $z \in [q,x]$ satisfy $d(z,q) = \theta d(x,q)$. For w in the convex set,

$$d(z,w) \geq d(x,w) - d(x,z) \geq \theta d(x,q) = d(z,q).$$

Uniqueness gives $Pz = q$. Projection non-expansiveness and (18), applied to the midpoints of $[x,Px]$ and $[y,Py]$, now imply

$$d(x,y)^2 - d(Px,Py)^2 \geq \tfrac{1}{2}\big(d(x,\mathcal{F}) - d(y,\mathcal{F})\big)^2.$$

The function $\psi(s) = \ell(s/\Delta) - s^2/(2\Delta^2)$ is convex, since its second derivative on its finite domain is $\Delta^{-2}\tan^2(s/\Delta)$. Decompose the kinetic cost as

$$\mathcal{K}_{\Delta,W}(h,k) = \frac{W}{4\Delta^2} d_2(h,k)^2 + \frac{W}{2}\sum_i \psi(\sigma_i(h,k)).$$

Use the preceding distance inequality for the quadratic term and (21) for the remainder. □

When applying these lemmas to a polygon, we project its nodes and then join the projected nodes by geodesics. We do not require projection to commute with geodesic interpolation. Fixed endpoint comparisons are valid only when the endpoint norms belong to $\mathcal{F}$, unless the changed endpoints are separately restored.

2.4. **Common bases and order retractions.** The next finite-dimensional observation permits spectral comparisons for operations on a pair of norms whose apartments need not agree.

Lemma 2.5. *Let $h(s)$ and $k(s)$, $0 \leq s \leq 1$, be split geodesics, each in its own fixed apartment. There is a finite subdivision on each closed piece of which both curves are diagonal in one fixed basis, with affine weights. The same statement holds with real-analytic weights defined on a neighborhood of $[0,1]$, with analytic weights in the common bases.*

Proof. Write the two fixed bases as $e, f = eM$, with weights $a_i(s), c_j(s)$. At a given parameter choose a nonzero pivot maximizing

$$\log|M_{ij}| + a_i(s) - c_j(s).$$

Its column inequalities say $|M_{kj}/M_{ij}|e^{a_k} \leq e^{a_i}$. Thus replacing e_i by $e_i + \sum_{k\neq i}(M_{kj}/M_{ij})e_k$ is an isometric triangular change of the h-basis, and makes $f_j = M_{ij}e_i^{\mathrm{new}}$. The row inequalities similarly give $|M_{il}/M_{ij}|e^{c_j} \leq e^{c_l}$; replacing f_l by $f_l - (M_{il}/M_{ij})f_j$ for $l \neq j$ is an isometric change of the k-basis and eliminates the pivot row. The inverses satisfy the same inequalities. Continue on the constant invertible Schur complement.

There are finitely many complete pivot sequences. Each produces constant basis matrices; its validity is a finite list of inequalities in the original weights plus constants. For affine weights each validity set is an interval, possibly empty or a point. Their union covers $[0,1]$, and subdivision at the finitely many endpoints gives the assertion. At completion the two final bases differ by a permutation and a diagonal matrix, so the new weights differ from the original weights only by fixed constants and permutation.

For analytic weights the differences defining validity are analytic on a neighborhood of the closed interval. Each is either identically zero or has finitely many zeros there. Subdivide at these zeros for all the finitely many pivot sequences. Validity is constant on each open piece

and extends to its endpoints by non-strict inequalities. This proves the analytic assertion as well. Ties require no separate choice of a limiting basis. □

For two norms define their maximum and infimal convolution by

$$(h \vee k)(z) = \max\{h(z), k(z)\}, \qquad (h \wedge k)(z) = \inf_{z=x+y} \max\{h(x), k(y)\}.$$

The latter is a norm: equivalence to a fixed finite-dimensional norm gives a positive lower bound on nonzero vectors, and approximate minimizing decompositions give the ultrametric triangle inequality. In a common orthogonal basis these operations are coordinatewise maximum and minimum of logarithmic weights. Both are jointly non-expansive for the maximum d_∞ distance, as is clear from multiplicative comparison of their defining formulas.

For an arbitrary pair $H = (h_A, h_C)$ define

$$\Pi H = \big(\mathrm{mid}(h_A, h_A \wedge h_C), \mathrm{mid}(h_C, h_A \vee h_C)\big). \tag{23}$$

For an ordered pair and $s \geq 0$, also define

$$\begin{aligned} \mathsf{S}_s H &= \big(\mathrm{mid}(h_A, h_C \wedge e^s h_A), \mathrm{mid}(h_C, h_A \vee e^{-s} h_C)\big), \\ \mathsf{C}_s H &= \big(\mathrm{mid}(h_A, h_A \vee e^{-s} h_C), \mathrm{mid}(h_C, h_C \wedge e^s h_A)\big). \end{aligned} \tag{24}$$

These formulas are intrinsic. In a common basis with weight pair (a_i, c_i), Π leaves the pair unchanged when $a_i \leq c_i$ and otherwise replaces it by its common average. For the two ordered operations, put $m_i = (a_i + c_i)/2$ and $g_i = c_i - a_i \geq 0$. They retain m_i and replace g_i, respectively, by

$$(g_i - s)_+ \quad \text{and} \quad \min\{g_i, s\}. \tag{25}$$

Lemma 2.6. *The map Π retracts all norm pairs onto $\mathcal{D}$. The maps $\mathsf{S}_s, \mathsf{C}_s$ preserve $\mathcal{D}$ and do not increase its determinant gap. All three preserve $d(h_A) + d(h_C)$, preserve splitness, and are continuous in the maximum d_∞ metric. For any one of these maps T, with s fixed where applicable,*

$$\sum_{v,i} \phi\big(\sigma_i((TH)_v, (TH')_v)\big) \leq \sum_{v,i} \phi\big(\sigma_i(h_v, h'_v)\big) \tag{26}$$

for every lower-semicontinuous convex ϕ. The same scalar function is used at both vertices.

Proof. The pointwise assertions follow from the common-basis formulas. Joint d_∞ continuity follows from the maximum, infimal convolution and canonical midpoint formulas, and extends the assertions to all norms by split approximation. Ordered pairs can be approximated by split ordered pairs: after sufficiently close split approximation, a vanishing opposite scalar shift restores order.

For the cost inequality first join H to H' by a product geodesic on $[0, 1]$. By Lemma 2.5 it has finitely many fixed common-basis pieces. Subdivide further where a signed gap is zero, or where an ordered gap equals s. On every resulting piece the transformed weights are affine and their derivatives either stay unchanged or are replaced by their pairwise average. Jensen's inequality gives

$$\sum_{v,i} \phi((Th)'_{vi}) \leq \sum_{v,i} \phi(h'_{vi}).$$

The right-hand side is the constant endpoint spectral cost of the original product geodesic. Apply the spectral triangle majorization and Jensen's inequality to the transformed finite path. Its endpoint cost is at most the integral of its derivative cost, proving (26) for split inputs. For all-norm inputs first pass to the limit for finite continuous convex ϕ. A general

lower-semicontinuous convex function is the increasing supremum of finite maxima of affine minorants; this gives the extended-valued assertion without assuming continuity at a barrier. □

In particular the three maps contract the total log-cosine kinetic cost between two nodes. The parameter s in this statement is constant. A time-dependent shrink introduces additional velocities, which will be estimated separately in the proof of finite incoming contact.

3. The renormalized variational problem

Set $q = a/2$. We first define an exact action on geodesic polygons, then relax it in the uniform topology. The construction will not assume that a limiting norm is split.

3.1. **Exact polygonal action.** Let $I = [l, r]$ and fix ordered endpoint pairs. An ordered polygon c is a continuous curve in $\mathcal{D}$ which is an affinely parametrized product geodesic on each interval of a finite partition $l = u_0 < \cdots < u_N = r$. Its nodes may be arbitrary norms. Write

$$\Delta_j = u_{j+1} - u_j, \qquad W_j = \int_{u_j}^{u_{j+1}} R(u)\,\mathrm{d}u, \qquad g(c) = d(c_C) - d(c_A) \geq 0.$$

Its action is

$$E_I(c) = \sum_j \left\{ \frac{W_j}{2} \sum_{v=A,C} \sum_i \ell\left(\frac{\sigma_i(c_v(u_j), c_v(u_{j+1}))}{\Delta_j} \right) + q \int_{u_j}^{u_{j+1}} R(u) g(c(u))\,\mathrm{d}u \right\}. \tag{27}$$

The determinant gap is affine on a geodesic. Its integral on a segment is therefore a positive weighted sum of the two endpoint gaps. Refining a partition without changing its curve leaves the action exactly unchanged: the relative spectra on subsegments scale with length and the integrals add.

Proposition 3.1. *On every feasible fixed partition, (27) has a unique minimum among all ordered nodes with the fixed endpoints. If the endpoints are split over K, all nodes of this minimum are split over K.*

Proof. The objective is nonnegative, lower semicontinuous and convex by Lemma 2.1, and the potential is affine in the nodes. Finite cost implies

$$d_\infty(c_v(u_j), c_v(u_{j+1})) < \tfrac{\pi}{2}\Delta_j.$$

The triangle inequality from a fixed endpoint thus bounds every node of every finite sublevel set. A proper lower-semicontinuous convex function with bounded nonempty sublevel sets on a complete CAT(0) space attains its infimum. Here one can see attainment directly: project a fixed origin to the nested closed convex sets with objective at most the infimum plus $1/m$. The CAT(0) projection inequality gives

$$d(p_m, p_k)^2 \leq d(o, p_k)^2 - d(o, p_m)^2 \qquad (k \geq m).$$

The right-hand side tends to zero, so the projections converge to a point in every sublevel set.

For uniqueness, compare two polygons c, c' on the same partition. Write b_{vj} for the distance between their v-norms at node j, and interpolate their midpoint nodes to obtain m.

Decompose $\ell(s/\Delta)$ as $s^2/(2\Delta^2)$ plus a convex function. Apply (18) to the quadratic part and spectral convexity to the remainder. The potential at the midpoint nodes is affine, giving

$$E_I(m) \le \frac{E_I(c) + E_I(c')}{2} - \frac{1}{16} \sum_{v,j} \frac{W_j}{\Delta_j^2} (b_{v,j+1} - b_{vj})^2. \tag{28}$$

If both curves minimize, every distance difference vanishes. The endpoint distances are zero, so every node distance is zero.

If the endpoints are split, apply Lemma 2.2 to the finite diagram of time nodes, with identity arrows between consecutive nodes and from C to A at each node. The horizontal spectra preserve the kinetic terms. The vertical spectra preserve order and the gap g, hence the exact potential. The endpoint norms are retained in full. A split tuple therefore has the same minimum value, and uniqueness identifies it with the minimum already constructed. □

Let $\mathcal{P}_I$ be the finite-action polygons with the fixed endpoints, and suppose this class is nonempty. Set $E_I^* = \inf_{c \in \mathcal{P}_I} E_I(c)$. On continuous ordered curves with these endpoints define

$$\overline{E}_I(c) = \inf \left\{ \liminf_m E_I(c_m) : c_m \in \mathcal{P}_I,\ c_m \longrightarrow c \text{ uniformly} \right\}, \tag{29}$$

with the value $+\infty$ when there is no such sequence.

Proposition 3.2. *The functional (29) has a unique minimum c_I, with value E_I^*. Every polygonal minimizing sequence converges uniformly to c_I. Its vertex curves are $\pi/2$-Lipschitz for d_∞ and $\sqrt{n}\pi/2$-Lipschitz for d_2. If the endpoints are split, a minimizing sequence can be chosen with all its nodes split.*

Proof. For two finite-action polygons use a common refinement. Since $W_j \ge \Delta_j$, (28), the zero endpoint distances and Cauchy–Schwarz give

$$\sup_{u \in I, v} d_2(c_v(u), c_v'(u))^2 \le 16|I| \left(\frac{E_I(c) + E_I(c')}{2} - E_I^* \right). \tag{30}$$

The node bound extends to every time by Busemann convexity of the geodesics. Thus any minimizing sequence is uniformly Cauchy in the complete space of continuous ordered curves. The same inequality identifies the limits of any two such sequences. The relaxation attains value E_I^* at this limit, and a diagonal sequence in the definition of the relaxation proves uniqueness. The barrier bounds the speed of every finite-action polygon on each segment; the stated Lipschitz constants pass to the limit. For split endpoints, replace each polygon by the minimum on its own partition. Its action decreases and its nodes are split by Proposition 3.1. □

The final replacement in the proof is an energy comparison. It need not approximate the original polygon at its interior nodes. It also does not imply that the uniform limit is split.

Lemma 3.3. *Suppose c has finitely many fixed-apartment pieces on I, with C^1 weights up to their endpoints and all their derivatives in a compact subinterval of $(-\pi/2, \pi/2)$. Then*

$$\overline{E}_I(c) = \int_I R(u) \left\{ \frac{1}{2} \sum_{v,i} \ell(w_{vi}'(u)) + q g(c(u)) \right\} \mathrm{d}u. \tag{31}$$

Proof. For an arbitrary norm polygon on a subinterval of length D, (17) and Jensen's inequality imply

$$D\sum_i \ell\left(\frac{\sigma_i(h(l),h(r))}{D}\right) \le \sum_j \Delta_j \sum_i \ell\left(\frac{\sigma_{ji}}{\Delta_j}\right). \tag{32}$$

Consider polygons converging uniformly to c. For a fixed subdivision of I, multiply (32) on each interval by its minimum value of R and sum, with the factor $1/2$ and both vertices. Spectral continuity and lower semicontinuity pass this endpoint lower bound to the limit. The potential integrals converge because determinant logarithms are Lipschitz on the compact interval. Refine the subdivision inside the finitely many apartments of c. Its endpoint spectral quotients are the increments of its weights, up to permutation, and converge uniformly to their derivatives on each piece. The strict velocity margin gives convergence of the lower bounds to the right-hand side of (31). This proves the lower inequality. This argument also permits converging, rather than identically fixed, endpoints in the approximating sequence.

Conversely, join the sampled values of c on refining subdivisions by product geodesics. Within each piece these are the linear interpolants of the weights. They remain ordered, keep the strict velocity margin, converge uniformly, and their exact actions converge to the integral. This proves the upper inequality. □

3.2. **Whole-line comparison curves.** Fix the full boundary triple and set

$$r_A(u) = e^{-F(u)}h_A^+, \qquad r_C(u) = e^{F(u)}h_C^+ \qquad (u \ge 0).$$

For sufficiently large u this pair is ordered. Let $\mathcal{Q}$ be the continuous ordered split curves for which, for some $U > 0$,

$$c_A = c_C = h_B \quad (u \le -U), \qquad c_v = r_v \quad (u \ge U), \tag{33}$$

and whose restriction to $[-U,U]$ has finitely many fixed-apartment C^1 pieces with strict velocity margins. Derivative jumps at the finitely many joins are allowed in this comparison class.

This class is nonempty for every triple. Since $F(U) = (\pi/3)U + O(1)$, the d_∞ distances of $r_A(U), r_C(U)$ from h_B are at most $(\pi/3)U + O(1)$. On $[-U,U]$ the product geodesic from (h_B,h_B) to $(r_A(U),r_C(U))$ therefore has spectral speeds less than $\pi/4$ for large U. It is ordered by geodesic convexity. Appending the reference tails gives an element of $\mathcal{Q}$.

Define the Bregman difference

$$D_\ell(v;s) = \ell(v) - \ell(s) - \ell'(s)(v-s), \qquad s_A(u) = -\beta(u), \quad s_C(u) = \beta(u).$$

On $\mathcal{Q}$ define

$$\begin{aligned} J(c) = &\int_{-\infty}^{0} R(u)\left\{\frac{1}{2}\sum_{v,i}\ell(w'_{vi}(u)) + qg(c(u))\right\}\mathrm{d}u \\ &+ \frac{1}{2}\int_0^\infty R(u)\sum_{v,i} D_\ell(w'_{vi}(u);s_v(u))\,\mathrm{d}u. \end{aligned} \tag{34}$$

All integrands are nonnegative, and they vanish outside a compact interval. The velocity multisets and determinant gap are intrinsic, so the value does not depend on the apartment presentations.

The renormalization has an exact finite-interval identity. If J_U denotes the two integrals in (34) restricted to $[-U,U]$, then for every finite regular ordered curve on that interval,

$$J_U(c) = E_{[-U,U]}(c) - \frac{a\sinh U}{2} g(c(U)) + C_n(U), \tag{35}$$

where the differential action is used and

$$C_n(U) = n\int_0^U R(u)\{\beta(u)\tan\beta(u) - \ell(\beta(u))\}\,\mathrm{d}u.$$

Indeed, $R\ell'(s_C) = a\sinh u = -R\ell'(s_A)$ and $(a\sinh u)' = aR$. Integration by parts cancels the potential on the positive half-line. The term at zero vanishes, and the interior boundary terms cancel by continuity of determinant norms. No reference endpoint is needed for this truncated identity. For a curve in $\mathcal{Q}$ and a cutoff containing its nonreference part, $J_U(c) = J(c)$. The endpoint term in (35) then depends only on the fixed boundary triple and U.

3.3. **Existence, uniqueness and the ends.** Let $J_* = \inf_{c\in\mathcal{Q}} J(c)$. Fix $c_0 \in \mathcal{Q}$ and take the uniform closure $\mathcal{Y}$ of $\mathcal{Q}$ among continuous ordered all-norm curves at finite uniform distance from c_0. Use $\sup_u \max_v d_2$ for this distance. The ambient space, and hence $\mathcal{Y}$, is complete. Define

$$\overline{J}(c) = \inf\{\liminf_m J(c_m) : c_m \in \mathcal{Q},\ c_m \to c \text{ uniformly}\}. \tag{36}$$

Theorem 3.4. *The functional $\overline{J}$ has a unique minimum c_* on $\mathcal{Y}$, with value J_*. Every minimizing sequence in $\mathcal{Q}$ converges uniformly to c_*. Its vertex curves are globally $\pi/2$-Lipschitz in d_∞. It has the full normalized limits (8) and is equivariant under simultaneous $\mathrm{GL}_n(K)$ change of basis.*

Proof. Take $c, c' \in \mathcal{Q}$ and a common cutoff U. By Lemma 3.3, they can be approximated on $[-U,U]$ by ordered split polygons on common refinements, with exact endpoints, uniform convergence and convergence of action. Attach the common reference tails. For two such polygons retain the coefficients W_j in (28). The midpoint polygon, with reference tails, belongs to $\mathcal{Q}$. By (35), its action is at least J_* and the endpoint corrections cancel. Moreover,

$$\sum_j \frac{\Delta_j^2}{W_j} \le \int_{-U}^{U} \frac{\mathrm{d}u}{R(u)} \le \pi.$$

Weighted Cauchy–Schwarz, starting from the zero endpoint distances, and Busemann convexity on each segment therefore give, after recovery,

$$\sup_{u,v} d_2(c_v(u), c_v'(u))^2 \le 16\pi \left\{ \frac{J(c) + J(c')}{2} - J_* \right\}. \tag{37}$$

Thus every minimizing sequence is uniformly Cauchy. Completeness and the definition of $\mathcal{Y}$ give a limit. The same inequality identifies all such limits. A diagonal sequence for (36) proves that its minimum value is J_* and that the minimum is unique. The Lipschitz bound follows from the barrier on every regular piece and passes to the uniform limit.

Since $\ell'' \ge 1$, one has $\ell(v) \ge v^2/2$ and $D_\ell(v;s) \ge (v-s)^2/2$. On a regular apartment piece the metric speed is the Euclidean norm of the weight derivatives. Integrate this speed from the constant negative tail, or integrate the speed of the scalar-normalized curve from

the positive reference tail. Weighted Cauchy–Schwarz gives, for $c \in \mathcal{Q}$,

$$\begin{aligned} d_2(c_v(u), h_B)^2 &\le 4J(c) \int_{-\infty}^{u} \operatorname{sech} s \, \mathrm{d}s \quad (u \le 0), \\ d_2(e^{-\varepsilon_v F(u)} c_v(u), h_v^+)^2 &\le 4J(c) \int_{u}^{\infty} \operatorname{sech} s \, \mathrm{d}s \quad (u \ge 0), \end{aligned} \tag{38}$$

where $\varepsilon_A = -1$ and $\varepsilon_C = 1$. These estimates pass to a minimizing sequence with $J(c)$ replaced by J_*. They prove the full norm limits, initially with rate $O(e^{-|u|/2}) = O(|t|^{-1/2})$. Finally, simultaneous change of basis preserves order, relative spectra, the determinant gap and the reference curves. It preserves J and its relaxed domain. Uniqueness proves equivariance. □

At this stage c_* is an all-norm curve; splitness and harmonicity have not been used to obtain its uniqueness. Both will be proved below, together with the stronger tail rates.

Proposition 3.5. *For large U, let c_U be the exact-endpoint relaxed minimum on $[-U, U]$ with endpoints (h_B, h_B) and $(r_A(U), r_C(U))$, extended by the reference tails. Put*

$$j(U) = E^*_{[-U,U]} - \frac{a \sinh U}{2} g(r_A(U), r_C(U)) + C_n(U).$$

Then $j(U)$ decreases to J_*, $c_U \in \mathcal{Y}$, and

$$\sup_{u,v} d_2((c_U)_v(u), (c_*)_v(u))^2 \le 8\pi\{j(U) - J_*\}. \tag{39}$$

Proof. Let $\mathcal{Q}_U$ be the curves in $\mathcal{Q}$ with the exact reference tails outside $[-U, U]$. The classes increase with U and their union is $\mathcal{Q}$. Regular-curve recovery shows that $j(U) = \inf_{\mathcal{Q}_U} J$. In the reverse comparison needed for this equality, an arbitrary all-norm polygon can be replaced by its split minimum on the same partition, with the same split endpoints and no larger action. Hence $j(U) \downarrow J_*$.

The split polygonal minimizing sequence for c_U, extended by its reference tails, lies in $\mathcal{Q}_U$, converges uniformly and has action tending to $j(U)$. This puts c_U in $\mathcal{Y}$. Apply (37) to that sequence and a whole-line minimizing sequence, then pass to both limits. □

Proposition 3.6. *Canonical scalar extension identifies the whole-line minima over K and over L:*

$$J^*_K = J^*_L, \qquad c_{*,L}(u) = (c_{*,K}(u))_L \quad \text{for every } u \in \mathbb{R}.$$

Proof. Extension preserves relative spectra, determinant gaps and regular apartment weights, so it carries $\mathcal{Q}_K$ into $\mathcal{Q}_L$ with the same action. This proves $J^*_L \le J^*_K$. Conversely, approximate a curve of $\mathcal{Q}_L$ by polygons on a fixed cutoff interval, with its exact split K endpoint norms extended to L. The finite formula in the proof of Lemma 2.2 applies directly to this L-diagram, using the fixed K endpoint matrices and real monomials in K as parameters. It gives a K-polygon preserving all horizontal and vertical profiles. Its order, exact potential, kinetic action and full endpoints are therefore preserved. Append the K reference tails and pass to the infimum. This proves the reverse inequality.

A minimizing sequence over K extends isometrically to a minimizing sequence over L. Its uniform limit is the canonical extension of $c_{*,K}$. The uniqueness in Theorem 3.4, now over L, identifies that limit with $c_{*,L}$. □

4. Local regularity of the minimum

We continue to regard the minimum as a curve in the complete all-norm space. Its first regularity properties are metric and spectral; finite apartments will be obtained later.

4.1. **Time allocation and strict velocity margins.** The barrier alone bounds speeds by $\pi/2$. To make local variations, we need a strict bound, independent of the number of polygonal segments.

Lemma 4.1. *Fix a compact interval $I = [l, r]$, its length $H > 0$, and an ordered polygon of action at most E_0. Its time allocation can be changed, keeping the ordered norm nodes and endpoints, so that its action does not increase and every spectral velocity has absolute value at most $b < \pi/2$. The constant depends only on I, E_0, n, and q.*

Proof. Minimize the exact action over the compact simplex of ordered node times. A zero-duration segment has infinite cost unless its endpoints coincide; in that case its cost is zero. Every nonzero edge has a strictly positive minimum feasible duration and its cost diverges at the barrier. Constant segments have continuous potential costs as their duration vanishes. Thus the extended objective is lower semicontinuous and attains a finite minimum. Delete zero-duration segments. All remaining durations are positive, so the derivatives with respect to interior node times vanish.

For segment j put $\overline{R}_j = W_j/\Delta_j$, and let v_{ji} range over all $2n$ spectral velocities. Define

$$L_j = \sum_i \ell(v_{ji}), \qquad H_j = \sum_i v_{ji} \tan v_{ji}, \qquad Q_j = \tfrac{1}{2}\overline{R}_j(H_j - L_j).$$

If g_j is the node determinant gap, then $\kappa_j = (g_{j+1} - g_j)/\Delta_j$ is the difference of the two velocity sums, hence $|\kappa_j| \le n\pi$. Set

$$b_{Lj} = \int_{u_j}^{u_{j+1}} R(u)\frac{u_{j+1} - u}{\Delta_j}\,\mathrm{d}u, \qquad b_{Rj} = \int_{u_j}^{u_{j+1}} R(u)\frac{u - u_j}{\Delta_j}\,\mathrm{d}u.$$

Differentiating the exact kinetic and potential integrals at an interior node, whose norm is held fixed, gives

$$\begin{aligned} Q_j - Q_{j-1} = {} & \tfrac{1}{2}(R(u_j) - \overline{R}_j)L_j - \tfrac{1}{2}(R(u_j) - \overline{R}_{j-1})L_{j-1} \\ & + q(\kappa_{j-1}b_{R,j-1} + \kappa_j b_{Lj}). \end{aligned} \tag{40}$$

For example, the right-end kinetic derivative is $R(u_{j+1})L_j/2 - \overline{R}_j H_j/2$, and the right-end potential derivative is $qR(u_{j+1})g_{j+1} - q\kappa_j b_{Rj}$. The corresponding left-end derivatives cancel the endpoint potential terms and give (40).

Let $M = \max_I R$ and $S = \max_I |R'|$. Since the potential is nonnegative and $R \ge 1$, $\sum_j \Delta_j L_j \le 2E_0$. Using $|R(u_j) - \overline{R}_j| \le S\Delta_j$ and the analogous bound on the preceding segment gives

$$\sum_j |Q_j - Q_{j-1}| \le 2SE_0 + 2qn\pi MH. \tag{41}$$

Some segment has $L_{j_0} \le 2E_0/H$. Every velocity on it has absolute value at most $b_0 = \arccos(e^{-2E_0/H})$, so $Q_{j_0} \le Mnb_0 \tan b_0$. Thus all Q_j have a bound independent of the number of segments. The function

$$f(v) = v \tan v - \ell(v)$$

is even, nonnegative and strictly increasing for $v > 0$, and tends to infinity as $v \uparrow \pi/2$. Since $Q_j \ge \frac{1}{2}f(v_{ji})$, the asserted strict bound follows. □

Proposition 4.2. *On every compact interval the vertex curves of the whole-line minimum are b-Lipschitz in d_∞ for some $b < \pi/2$. There is a global minimizing sequence with polygonal restriction to that interval and the same strict spectral velocity bound there.*

Proof. Start with split polygonal almost-minima on expanding cutoff intervals, attached to their reference tails. Proposition 3.5 and regular-curve recovery make their J values tend to J_*, so they converge uniformly to c_*. For a fixed symmetric compact interval $I = [-U, U]$, the truncated identity (35) bounds their local E_I values: their truncated nonnegative J_I integrals are bounded by J, and their endpoint determinant gaps converge. Apply Lemma 4.1 only on I, leaving its endpoint norms and the exterior unchanged. The change in J is the change in E_I, since all boundary terms cancel. The new curves are still in $\mathcal{Q}$ and still minimize asymptotically. They therefore have the same uniform limit by (37). Their uniform strict bound on I passes to that limit. An arbitrary compact interval lies inside such an I. □

4.2. **Exact-endpoint local minimality.** The endpoints of a compact restriction of c_* might still be nonsplit. The following repair allows local competitors with those exact endpoints.

Lemma 4.3. *Suppose ordered polygons p_m on I converge uniformly to c and have a common strict spectral velocity margin. They can be replaced by ordered all-norm polygons $\widehat{p}_m$ with the exact endpoints of c, such that $\widehat{p}_m \to c$ uniformly and $E_I(\widehat{p}_m) - E_I(p_m) \to 0$. Conversely, every fixed finite-action polygon p with the endpoints of c has polygonal replacements z_m with the endpoints of p_m and $E_I(z_m) \to E_I(p)$.*

Proof. Let δ_m be the maximum endpoint d_2 error. Choose $\eta_m \downarrow 0$ with $\delta_m/\eta_m \to 0$ and $2\eta_m < |I|$. Add product-geodesic bridges of length η_m at the two ends, and compress p_m affinely into the middle interval. Ordered endpoints give ordered bridges. The middle velocities are multiplied by $1 + O(\eta_m)$ and retain a fixed strict margin; the bridge velocities are bounded by δ_m/η_m.

Change variables on the middle interval. Its Jacobian, time weight and velocity scaling change by $O(\eta_m)$, while ℓ and ℓ' are uniformly bounded on the strict margin. Thus its kinetic change is $O(\eta_m)$, independently of the number of segments. Uniformly bounded determinant gaps give the same potential bound. Each bridge has kinetic cost $O(\delta_m^2/\eta_m) = o(1)$ and potential cost $O(\eta_m)$. The uniform Lipschitz bound and small time displacement prove uniform convergence. Reverse the bridges and compress the fixed polygon p to prove the converse assertion. Its own finite partition supplies the strict margin needed for the same estimates. □

Proposition 4.4. *Every compact restriction of c_* is the unique exact-endpoint local relaxed minimum of (29).*

Proof. Use the sequence from Proposition 4.2 on I. The first endpoint repair gives $E_I^* \le \liminf_m E_I(p_m|I)$. The local problem is feasible, since the strict metric speed bound makes the endpoint chord have finite cost. Take any local polygon p with the exact limiting endpoints. The converse repair gives z_m with the split endpoints of $p_m|I$ and $E_I(z_m) \to E_I(p)$.

The intermediate nodes of z_m need not be split. Replace it by the minimum on its finite partition. Proposition 3.1 makes this a split polygon $\widetilde{z}_m$ with the same endpoints and no larger action. It can now be inserted into p_m while retaining membership in $\mathcal{Q}$. Minimality of the whole-line infimum and cancellation of the finite boundary terms give

$$E_I(p_m|I) \le E_I(\widetilde{z}_m) + J(p_m) - J_* \le E_I(z_m) + J(p_m) - J_*.$$

Taking the upper limit and then the infimum over p proves $E_I(p_m|I) \to E_I^*$. Their exact-endpoint repairs have the same limiting energy and converge uniformly to $c_*|I$. Proposition 3.2 identifies this curve with the unique local minimum. □

4.3. **An ordered kinetic comparison.** Work over L, where all endpoint norms are split. Fix $I = [l, r]$ and ordered endpoint pairs with $d_\infty(h_v(l), h_v(r)) < \pi|I|/2$. At each vertex choose a common endpoint basis, with weights a_{vi}, b_{vi}. There is a unique real p_{vi} satisfying

$$\int_l^r \arctan(p_{vi} \operatorname{sech} u)\, \mathrm{d}u = b_{vi} - a_{vi}. \tag{42}$$

The left side is strictly increasing from $-\pi|I|/2$ to $\pi|I|/2$. Define a split curve k in these endpoint apartments by

$$w_{vi}(l) = a_{vi}, \qquad w'_{vi}(u) = \arctan(p_{vi} \operatorname{sech} u), \qquad R(u) \tan w'_{vi}(u) = p_{vi}. \tag{43}$$

Lemma 4.5. *The curve k minimizes the pure kinetic action uniquely in its uniform polygonal relaxation with these endpoints. It is ordered. If γ is the affine endpoint chord, then*

$$\sup_{u \in I} d_2(k_v(u), \gamma_v(u)) \le \frac{\sqrt{n}}{16} |I|^2. \tag{44}$$

Proof. Project a comparison polygon separately at the two vertices onto their respective endpoint apartments using Lemma 2.3. This fixes the endpoints and decreases the kinetic cost. In either apartment, the supporting inequality for ℓ and the constant momenta in (43) give

$$\int_I \frac{R}{2} \sum_i \ell(z'_i)\, \mathrm{d}u \ge \int_I \frac{R}{2} \sum_i \ell(w'_i)\, \mathrm{d}u + \frac{1}{2} \sum_i p_i \int_I (z'_i - w'_i)\, \mathrm{d}u.$$

The final term vanishes by the endpoints. This proves the lower bound against all polygons and, by recovery, identifies the kinetic infimum with $K(k)$.

The midpoint calculation without potential gives, for two exact-endpoint polygons z, z',

$$\sup_{u,v} d_2(z_v(u), z'_v(u))^2 \le 16 B_I \left\{ \frac{K(z) + K(z')}{2} - K(k) \right\}, \qquad B_I = \int_I \operatorname{sech} u\, \mathrm{d}u. \tag{45}$$

This follows from (28) and $\sum \Delta_j^2 / W_j \le B_I$. It proves uniqueness of the relaxed kinetic minimum.

Sample k on refining partitions and apply Π to its nodes. The projected polygons have the same ordered endpoints and no larger kinetic cost by Lemma 2.6. Their costs still converge to $K(k)$. Inequality (45) forces them to converge uniformly to k, and closedness of $\mathcal{D}$ makes k ordered. This argument does not identify a projected geodesic with the geodesic joining its projected endpoints.

Finally, differentiating (43) gives

$$w''_{vi} = -\tanh u \, \sin(w'_{vi}) \cos(w'_{vi}), \qquad |w''_{vi}| \le \tfrac{1}{2}.$$

The scalar interpolation error from the affine chord is at most $|I|^2/16$ in each coordinate. Summing squares proves (44). □

Proposition 4.6. *For $I = [l, r]$, let γ be the affine endpoint chord of $c_*|I$. With $M = \max_I R$ and $m = \min_I R$,*

$$\sup_{u \in I} d_2((c_*)_v(u), \gamma_v(u)) \le \sqrt{n} \left(16 q \frac{M}{m} + \frac{1}{16} \right) |I|^2. \tag{46}$$

In particular, for $|I| \le 1$ the coefficient is bounded by $C = \sqrt{n}(16qe + 1/16)$, independently of the location of I and the boundary triple.

Proof. First work over L, using Proposition 3.6. The strict local margin makes the kinetic bridge k feasible for the endpoints of $c_*|I$. Put $V(c) = q\int_I Rg(c)\,\mathrm{d}u$ and $\overline{K}(c) = \overline{E}_I(c) - V(c)$. This subtracts a continuous potential from the relaxation; it does not presume a differential action for c_*. Exact-endpoint recovery and (45) give, for $D = \max_v \sup_I d_2((c_*)_v, k_v)$,

$$D^2 \le 8B_I\{\overline{K}(c_*) - K(k)\}.$$

Local minimality and the orderedness of k imply

$$0 \le \overline{K}(c_*) - K(k) \le V(k) - V(c_*).$$

The determinant logarithm is $\sqrt{n}$-Lipschitz for d_2. Thus $|g(k) - g(c_*)| \le 2\sqrt{n}D$, and

$$D^2 \le 16q\sqrt{n}B_I M|I|D.$$

Divide by D if it is nonzero and use $B_I \le |I|/m$. Combining with (44) proves (46). Since $|\mathrm{d}(\log R)/\mathrm{d}u| \le 1$, $M/m \le e^{|I|}$. Canonical extension is an isometry and preserves the endpoint chord, so the result also holds for the original K-norm curve. □

4.4. **Intrinsic spectral velocities.** For relative spectra we shall use the coordinatewise comparison

$$\big\|\sigma(h,k) - \sigma(h',k')\big\|_\infty \le d_\infty(h,h') + d_\infty(k,k'). \tag{47}$$

For split norms this follows by monotone comparison with scalar multiples in the min–max description of each relative spectral coordinate; see [BE21, Lemma 2.23]. Continuity extends it to all norms. This estimate is stronger than the partial-sum bound needed for earlier convexity arguments.

Proposition 4.7. *At each vertex, the oriented relative secants of c_* have a common forward and backward decreasing limit $v(u) \in \mathbb{R}^n$ at every time. For $0 < H \le 1$,*

$$\left\|\frac{\sigma(c_*(u), c_*(u+H))}{H} - v(u)\right\|_\infty \le 4CH, \tag{48}$$

with the same bound for the backward secant whose source is $u - H$ and target is u. The list v is locally Lipschitz, with constant $8C$. Its entries have a strict margin inside $(-\pi/2, \pi/2)$ on every compact interval. The determinant logarithm is $C^{1,1}$ locally and has derivative $\sum_i v_i$.

Proof. Fix a vertex and write $h(u)$ for its curve. Let $f_u(H) = \sigma(h(u), h(u+H))/H$. The midpoint case of (46), spectral scaling on geodesics and (47) give

$$\|f_u(H/2) - f_u(H)\|_\infty \le 2CH.$$

Summation on dyadic scales gives a limit and the error $4CH$. For $H/2 \le h \le H$, comparison with the chord at time $u + h$ gives $\|f_u(h) - f_u(H)\|_\infty \le 2CH$, identifying the same limit along arbitrary scales. The backward secants have the same construction. For $0 < H \le 1/2$, on $[u - H, u + H]$ the two half-chord spectra agree, and the actual midpoint differs by at most $4CH^2$. Thus the forward and backward secants differ by at most $8CH$, and their limits coincide.

For $s < t$ the forward secant from s to t and the backward secant at t are identical. Applying their error estimates gives $\|v(s) - v(t)\|_\infty \le 8C|s - t|$ for $t - s \le 1$. The strict margin follows from Proposition 4.2. Finally, the sum of the relative spectral coordinates is the determinant increment. Its two difference quotients tend to $\sum_i v_i(u)$ at every time, proving the last assertion. □

Proposition 4.8. *On every compact interval the relaxed action of the actual minimum is*

$$\overline{E}_I(c_*|I) = \int_I R(u) \left\{ \frac{1}{2} \sum_{v=A,C} \sum_i \ell(v_{vi}(u)) + qg(c_*(u)) \right\} \mathrm{d}u. \tag{49}$$

The spectral positions $x_{vi}(u) = R(u) \tan v_{vi}(u)$ are locally Lipschitz.

Proof. On a partition cell of length Δ, its relative spectral quotient differs by $O(\Delta)$ from the velocity list at the left endpoint, and hence at every time in that cell. The strict compact margin makes ℓ uniformly continuous on all these lists. The partition lower-bound argument in Lemma 3.3 therefore applies to any polygonal sequence converging to c_*, without a fixed apartment for the limit, and gives the lower bound in (49).

For recovery, sample c_* and interpolate canonically. All-norm nodes are permitted in the local relaxation. Order and the strict metric speed bound make these finite-action polygons, with exact endpoints. The same spectral-quotient estimate and the exact segment weights give convergence to the displayed integral. The potential converges uniformly as before. The final assertion follows from the locally Lipschitz velocity lists and their strict margins. □

The velocities and positions in these propositions are numerical lists. Their regularity alone does not identify embedded subspaces or make the positions straight. Those are the remaining algebraic and variational steps in the proof.

5. Algebraic tails and full incoming contact

In this section we prove that the two ends of the minimum have fixed apartments and the required algebraic labels. We work over L when choosing bases for norms whose splitness is not yet known. The scalar extension result identifies every conclusion about the resulting norms with a conclusion about the original minimum over K.

5.1. **The outgoing tail.**

Proposition 5.1. *For all sufficiently large u, each vertex norm of c_* lies in one fixed K-apartment. In its fixed basis its logarithmic weights satisfy*

$$R(u) \tan w'_{vi}(u) = \epsilon_v a \sinh u + p_{vi}, \qquad \epsilon_A = -1, \quad \epsilon_C = 1. \tag{50}$$

The identity arrow is strictly contracting. The associated positions are $x_{vi}(t) = \epsilon_v a t + p_{vi}$, and their labels are pure sums of A or C. The full normalized norm at each vertex converges to its prescribed limit at rate $O(e^{-u})$.

Proof. Fix $J_0 > J_*$. The tail estimate in the proof of Theorem 3.4 bounds the distance of every member p of a minimizing sequence with $J(p) \le J_0$ from the outgoing reference norms by

$$d_\infty(p_v(u), r_v(u)) \le \eta(u), \qquad \eta(u) = 2 \left(J_0 \int_u^\infty \operatorname{sech} s \, \mathrm{d}s \right)^{1/2}.$$

The same holds for c_*. Choose M with $h_A^+ \le e^M h_C^+$, and choose U so large that $2F(U) > M + 2\eta(U) + 1$. Any pair in these two tubes, at any $u \ge U$, satisfies

$$h_A \le e^{M + 2\eta(u) - 2F(u)} h_C \le e^{-1} h_C. \tag{51}$$

Over L, choose separately at each vertex a common basis for $(c_*)_v(U)$ and h_v^+. Let $\mathcal{A}_v$ be its apartment and P_v the projection of Lemma 2.3. It fixes both these norms and the entire reference curve r_v. Choose a global polygonal minimizing sequence p_m, with cutoffs $U_m \to \infty$,

whose velocities have a common strict margin on $[U, U+1]$. Project its nodes on $[U, U_m]$ by P_v, and interpolate on the same partition. Each projected node stays in its reference tube, so (51) makes the projected pairs ordered. Their interpolants are ordered as well. Although the two projections can differ, the tube estimate supplies the order required here. They preserve both determinant logarithms, hence all endpoint corrections in the renormalized action.

Let $D_m \geq 0$ be the decrease in $E_{[U,U_m]}$ under this node projection. The original norm at U need not be fixed by P_v. Its distance δ_m from the respective apartments tends to zero. Choose $\tau_m \downarrow 0$ with $\delta_m/\tau_m \to 0$, refine at $U+\tau_m$, and bridge from the original pair $p_m(U)$ to the projected pair at $U+\tau_m$. The bridge has velocities bounded by $b + \delta_m/\tau_m$ for a fixed $b < \pi/2$. It is ordered, fixes the original initial endpoint, and costs $O(\tau_m)$ more than the projected short interval. Leave the curve unchanged before U. This gives a member of $\mathcal{Q}_L$ with

$$J_* \leq J(\widetilde{p}_m) \leq J(p_m) - D_m + O(\tau_m).$$

Thus $D_m \to 0$.

Write $a_{vj} = d_2(p_{m,v}(u_j), \mathcal{A}_v)$. The departure estimate of Lemma 2.4 gives

$$D_m \geq \frac{1}{8} \sum_{v,j} \frac{W_j}{\Delta_j^2} (a_{v,j+1} - a_{vj})^2.$$

The departure at the final cutoff is zero. Weighted Cauchy–Schwarz, summed backwards from that endpoint, therefore gives

$$\sup_{u \geq U, v} d_2(p_{m,v}(u), \mathcal{A}_v)^2 \leq 8\pi D_m,$$

since $\sum \Delta_j^2/W_j \leq \int_U^\infty \operatorname{sech} u \, \mathrm{d}u \leq \pi$. Convexity of distance to an apartment extends the node bound to the segments. Closedness and uniform convergence imply that $(c_*)_v$ lies in $\mathcal{A}_v$ on the whole tail.

Its apartment weights are locally $C^{1,1}$. Indeed, in a fixed apartment the quadratic chord estimate is the Euclidean estimate for the unsorted weight coordinates. The dyadic argument of Proposition 4.7, applied to their coordinate increments, gives equal forward and backward derivatives and a locally Lipschitz derivative. The strict local velocity margin persists. Thus both the weights and their smooth compactly supported variations are covered by Lemma 3.3. The strict order (51) allows either sign of a small compactly supported variation of each individual weight. Local minimality gives, distributionally,

$$(R \tan w_{vi}')' = \epsilon_v a R.$$

Integration proves (50). Its right side makes the weights smooth. If b_{vi} is the logarithmic weight of h_v^+ in this basis, the full end limit and the elementary bound $|\arctan(x+y) - \arctan x| \leq |y|$ give

$$|w_{vi}(u) - \epsilon_v F(u) - b_{vi}| \leq |p_{vi}| \int_u^\infty \operatorname{sech} s \, \mathrm{d}s \leq 2|p_{vi}| e^{-u}. \tag{52}$$

The reduction of the strictly contracting arrow is zero. The straight branches consequently have pure A or C labels, subdivided at their finitely many intersections.

It remains to descend the fixed apartments. The next lemma applies to $e^{-\epsilon_v F(u)}(c_*)_v(u)$, whose limit is the split K-norm h_v^+. It gives a fixed K-basis on a shorter tail. Canonical extension then identifies its restriction with the original K-minimum. Weights differ only by fixed offsets, so their equations, labels and convergence rates are unchanged. □

Lemma 5.2. *Let h be a split K-norm. Suppose an L-norm curve $k(s)$ lies in one fixed apartment and converges to the extension of h as $s \to \infty$. On a sufficiently late tail, k is diagonal in one fixed K-basis.*

Proof. The limiting norm is diagonal in the same L-basis. Rescale its vectors by real powers of T to make that basis orthonormal for h. Its weights $z_i(s)$ for $k(s)$ then tend to zero. Choose an orthonormal K-basis f for h. The change-of-basis matrix P lies in $\mathrm{GL}_n(\mathcal{O}_L)$. Its residue is an invertible complex matrix; let P_0 be its constant lift. Write $P_0^{-1}P = I + Q$. There are only finitely many entries, so all nonzero entries of Q have valuation at least some $\delta > 0$. The inverse has the same property, by the adjugate formula modulo the ideal of elements of valuation at least δ.

For large s, the weight spread $\max z_i(s) - \min z_i(s)$ is less than δ. Both $I + Q$ and its inverse are then nonexpansive for the weighted maximum norm: every off-diagonal weighted entry has negative logarithm, and the diagonal entries are units. Thus replacing P by P_0 preserves that norm exactly. The basis fP_0 is the required fixed K-basis. This uses the fixed original apartment and the split limit; convergence alone would not give the conclusion. □

5.2. **The equation on a free interval.** The same scalar equation holds on every interval where the norm order is uniformly strict, without any assumption about its location relative to the ends.

Proposition 5.3. *Suppose $I = [l, r]$ satisfies $(c_*)_A \le e^{-\kappa}(c_*)_C$ for some $\kappa > 0$. Over L, choose at each vertex any common basis for the two endpoint norms. The minimum on I lies in these fixed apartments and satisfies (50) for suitable constants p_{vi}.*

Proof. Let a_{vi}, b_{vi} be the endpoint weights. There is a unique $p_{vi} \in \mathbb{R}$ such that

$$\int_l^r \arctan(\epsilon_v a \tanh u + p_{vi} \operatorname{sech} u)\, \mathrm{d}u = b_{vi} - a_{vi}.$$

Indeed the left side is strictly increasing with limits $\pm\pi|I|/2$, and the strict local margin puts the right side between them. Integrate these velocities from a_{vi} to obtain a smooth candidate z; initially it need not be ordered.

Project any ordered comparison polygon separately onto the two endpoint apartments. Its kinetic cost decreases and its determinant potential is unchanged. In those apartments, the supporting inequality for ℓ, integrated using $(R \tan w'_{vi})' = \epsilon_v a R$, proves that its action is at least the differential action $E(z)$. The fixed basis-volume constants in the determinant logarithms cancel. This comparison does not require the projected pair or z to be ordered. It proves $E_I^* \ge E(z)$.

Take exact-endpoint ordered polygonal almost-minima $p_m \to c_*|I$ and sample z on their refining partitions, obtaining z_m. For large m, $p_{m,A} \le e^{-\kappa/2} p_{m,C}$, whereas $z_{m,A} \le e^M z_{m,C}$ for some finite common $M \ge 0$. Choose $0 < \theta < 1$ with $-(1-\theta)\kappa/2 + \theta M \le 0$. Interpolate the two polygons' nodes by θ, then join those nodes geodesically. The resulting polygon r_m is ordered, has the original endpoints, and satisfies

$$E_I^* \le E(r_m) \le (1-\theta)E(p_m) + \theta E(z_m).$$

Taking limits proves $E(z) = E_I^*$ and $E(r_m) \to E_I^*$. Local stability makes $r_m \to c_*|I$. Its node-wise limit is $[c_*(u), z(u)]_\theta$, where brackets denote canonical geodesic interpolation. Refining meshes identify the same limit between nodes. Thus $c_* = [c_*, z]_\theta$, which forces $c_* = z$. □

5.3. **Small incoming speeds.** We need equality of the two incoming norms at a finite time. Contact of only the smallest relative gap would not suffice.

Lemma 5.4. *For $s \to -\infty$, the minimum and a global minimizing sequence can be chosen to have spectral metric speeds bounded by $O(\cosh(s)^{-1/4})$ on $I_s = [s-1, s+1]$. For the sequence this is a bound on every segment velocity.*

Proof. Refine the time-allocation estimate using a positive weight ρ on an interval of length H, with $0 < m \le \rho \le M$ and $|\rho'| \le S$. If the action is at most E_0, the optimized allocation satisfies

$$\sum_j \Delta_j L_j \le \frac{2E_0}{m}, \qquad \sum_j \Delta_j \sum_i v_{ji}^2 \le \frac{4E_0}{m}.$$

Consequently

$$\sum_j \Delta_j |\kappa_j| \le \sqrt{8nHE_0/m}.$$

In (40), the two potential endpoint weights of each segment sum to its integrated weight. Summing the kinetic and potential errors therefore gives the improved estimate

$$\sum_j |Q_j - Q_{j-1}| \le \frac{2SE_0}{m} + qM\sqrt{8nHE_0/m}.$$

Some segment has $L_j \le B = 2E_0/(mH)$. Put $b_0 = \arccos(e^{-B})$ and

$$A(E_0) = Mnb_0 \tan b_0 + \frac{2SE_0}{m} + qM\sqrt{8nHE_0/m}.$$

Every segment then has $Q_j \le A(E_0)$. Since $f(v) = v\tan v - \ell(v) \ge v^2/2$,

$$|v_{ji}| \le 2\sqrt{A(E_0)/m} = O(E_0^{1/4}).$$

The inverse of f also retains a strict $\pi/2$ margin when E_0 is not small. None of these bounds depends on the number of nodes.

Fix a global minimizing sequence with $J \le J_0$. On I_s, $s \le -2$, normalize the weight to $\rho_s(u) = R(u)/R(s)$. Then $e^{-1} \le \rho_s \le e$, $|\rho_s'| \le e$, and $H = 2$. Nonnegativity of the negative-half integrand bounds its normalized action by $E_0 = J_0/R(s)$. Reallocate time on I_s, keeping the endpoints and the exterior fixed. The global action decreases, so the sequence still converges uniformly to c_*. The preceding bound applies to the sequence and passes to the limit. □

Proposition 5.5. *There is a finite time U at which $(c_*)_A(U) = (c_*)_C(U)$.*

Proof. Take $I_s = [s-1, s+1]$ on the negative half-line, and let $\eta(u) = 1 - |u - s|$ there, extended by zero outside. Choose a polygonal minimizing sequence whose velocities on I_s are bounded by $b_s = O(R(s)^{-1/4})$. For any such polygon p, apply the time-dependent operation

$$\widetilde{p}(u) = S_{\varepsilon\eta(u)}(p_A(u), p_C(u)),$$

where S is the balanced operation in (24). This variation has the original full endpoints and reference tails. Lemma 2.5 gives a finite common-basis subdivision of every original segment. Refine at s and at all crossings of the affine gaps $g_i = c_i - a_i$ with $\varepsilon\eta$. On each remaining piece the new weights are

$$(a_i + \varepsilon\eta/2, c_i - \varepsilon\eta/2) \quad \text{if } g_i > \varepsilon\eta, \qquad ((a_i + c_i)/2, (a_i + c_i)/2) \quad \text{if } g_i < \varepsilon\eta.$$

They agree on equality intervals. Thus the new curve is a finite piecewise-apartment member of $\mathcal{Q}$, remains ordered, and has velocities at most $b_s + \varepsilon/2 < \pi/2$ for small ε.

Write $b = b_s$ and choose $b < b' < \pi/2$. Taylor's inequality, with $L_\ell = \sec^2 b'$, bounds the kinetic increase for each unclipped pair by

$$\frac{\varepsilon}{2}\tan b\,|\eta'| + \frac{L_\ell \varepsilon^2}{8}|\eta'|^2.$$

On a clipped pair Jensen's inequality gives no increase. The determinant gap drops by $\sum_i \min(g_i, \varepsilon\eta)$. If the largest gap is at least $\gamma > \varepsilon$ throughout I_s, this drop is at least $\varepsilon\eta$. Hence

$$J(\widetilde{p}) - J(p) \le -\varepsilon D_s(b) + \frac{nL_\ell\varepsilon^2}{8}\int_{I_s} R|\eta'|^2, \qquad D_s(b) = q\int_{I_s} R\eta - \frac{n}{2}\tan b\int_{I_s} R|\eta'|. \tag{53}$$

The elementary integrals are

$$\int_{I_s} R\eta = 2(\cosh 1 - 1)\cosh s, \qquad \int_{I_s} R|\eta'| = \int_{I_s} R|\eta'|^2 = 2\sinh 1\cosh s.$$

It follows that $D_s(b) > 0$ precisely when $n\tan b < \sqrt{3}\tanh(1/2)$. Choose s sufficiently negative for this inequality to hold.

If the two norms of c_* were different at every time in I_s, their continuous largest gap would have a positive minimum γ_*. Uniform convergence gives a largest gap at least $\gamma_*/2$ for all sufficiently late polygons. Fix $\varepsilon > 0$, independent of the polygon, below this bound and small enough that the quadratic term in (53) is less than $\varepsilon D_s(b)/2$. The transformed sequence then lowers J by a fixed positive amount, contradicting its convergence to J_*. Thus full equality occurs at a finite time. □

5.4. **The incoming tail.**

Proposition 5.6. *The full-contact set is $(-\infty, U_*]$ for some finite U_*. On a sufficiently negative tail the two norms coincide in one fixed K-apartment, with weights satisfying*

$$R(u)\tan w_i'(u) = p_i. \tag{54}$$

Their actual common norm converges to h_B at rate $O(e^u)$. The positions are horizontal lines $x_i(t) = p_i$, with pure semistable B labels.

Proof. Suppose full contact holds at U. Project a pair to its common midpoint, $P_D(h_A, h_C) = (m(h_A, h_C), m(h_A, h_C))$. Joint convexity of spectral costs shows that this decreases the total kinetic action of node polygons. It is ordered and removes the entire nonnegative determinant potential.

Choose a global polygonal minimizing sequence with a common strict velocity margin on $[U-1, U]$. Apply P_D to all nodes from its negative cutoff to U. Its action decrease D_m satisfies

$$D_m \ge q\int_{-U_m}^{U} R(u)g(p_m(u))\,\mathrm{d}u \ge 0.$$

The displacement of the final pair tends to zero. As in the outgoing proof, repair the last interval of length τ_m, with endpoint displacement $o(\tau_m)$, to the original pair $p_m(U)$. The bridge remains ordered and has a strict margin. It changes action by $O(\tau_m)$. Keeping the exterior fixed and using the truncated renormalization identity gives $J_* \le J(p_m) - D_m + O(\tau_m)$, even when $U > 0$. Thus $D_m \to 0$. Uniform convergence on every compact interval forces $g(c_*) = 0$ at all times before U. Ordered norms with zero determinant gap are equal, since all their relative gaps are nonnegative. Contact therefore propagates to the entire past.

Nonemptiness follows from Proposition 5.5; closedness and strict outgoing order identify its stated form.

Over L, choose a common basis for h_B and the common norm at U_*. Apply its apartment projection at both vertices on the negative tail of a minimizing sequence. It preserves order and determinant gaps. Repair the last endpoint as above. The projection loss tends to zero. Lemma 2.4, now summed forwards from zero departure at the negative reference endpoint, bounds the squared supremum departure by 8π times that loss. The limiting common norm consequently lies in this one apartment on $(-\infty, U_*]$.

As in the outgoing proof, the quadratic chord estimate makes the fixed-apartment weights locally $C^{1,1}$. Vary each weight equally at both vertices. The pair stays equal, its potential is zero, and both signs of the variation are admissible. Local minimality and Lemma 3.3 give $(R\tan w_i')' = 0$, proving (54). The weights are smooth, and the full incoming limit gives

$$|w_i(u) - \log h_B(e_i)| \le |p_i| \int_{-\infty}^{u} \operatorname{sech} s \, \mathrm{d}s \le 2|p_i| e^u.$$

Apply Lemma 5.2 with time reversed. On a shorter tail one fixed K-basis works at both vertices. The identity arrow is an isometry there, so equal momentum blocks have isomorphic reductions and carry precisely the claimed B labels. □

Both end estimates are $O(|t|^{-1})$ in the original coordinate. The unresolved part of the construction is now confined to a compact interval. The next section proves that it too has a finite algebraic description.

6. Finite algebraic regularity

We now pass from numerical spectral velocities to fixed algebraic subspaces. Work first over L, where every norm is split. No common apartment for the three boundary norms is assumed.

6.1. **Radial spectra and strict velocity gaps.**

Lemma 6.1. *Fix a reference norm o, and let $h(u)$ be either vertex curve of c_*. At every differentiability point of $\lambda(u) = \sigma(o, h(u))$, its derivative is a permutation of the intrinsic velocity list $v(u)$. For every k, the function $u \mapsto S_k(o, h(u))$ is locally semiconvex.*

Proof. We first give a uniform finite bound for the number of affine pieces of the radial spectrum along any apartment geodesic γ, of constant spectral velocity v. Apply the weighted elimination in Lemma 2.5 to the constant norm o and γ. A complete pivot sequence has at most $(n!)^2$ possibilities, since there are at most j^2 choices at residual size j. The validity set of each sequence is an interval, defined by affine weight inequalities. Their endpoints subdivide the parameter interval into at most $2(n!)^2 + 1$ open pieces. On each, one fixed common basis works. Its relative weights have derivatives that permute v. Ordering these affine weights adds at most $n(n-1)/2$ cuts per piece. Thus a sufficient uniform bound is

$$N(n) = \big(2(n!)^2 + 1\big)\big(n(n-1)/2 + 1\big).$$

This bound is independent of the basis, weights and interval length.

Fix a differentiability point u_0 of λ. Let γ_h be the chord of h on $[u_0 - h, u_0 + h]$, parametrized on that interval, and put

$$v_h = \frac{\sigma(h(u_0 - h), h(u_0 + h))}{2h}.$$

The quadratic chord estimate and (47) give $v_h \to v(u_0)$ and, uniformly for $|s| \le h$,

$$\sigma(o, \gamma_h(u_0 + s)) = \lambda(u_0 + s) + O(h^2) = \lambda(u_0) + s\lambda'(u_0) + o(h).$$

The left side has an affine piece of length at least $2h/N(n)$. Subtract the displayed equation at its endpoints and divide by that length. For a permutation P_h it follows that $P_h v_h = \lambda'(u_0) + o(1)$. Finiteness of the permutation group proves the assertion. This uses differentiability for one fixed reference o; no common full-measure set for all reference norms is needed.

The function $h \mapsto S_k(o, h)$ is geodesically convex by Lemma 2.1, and is k-Lipschitz for d_∞. Comparison with the midpoint of the endpoint chord therefore gives, locally uniformly,

$$S_k(o, h(u)) \le \tfrac{1}{2}\{S_k(o, h(u - h)) + S_k(o, h(u + h))\} + 4kCh^2.$$

Adding a sufficiently large quadratic makes this continuous function midpoint convex, hence convex. This proves local semiconvexity. □

Lemma 6.2. *Suppose $v_k > v_{k+1}$ throughout an open interval I. For every $s < t$ in I,*

$$S_k(h(s), h(t)) = \int_s^t \sum_{i \le k} v_i(u)\, \mathrm{d}u. \tag{55}$$

Proof. We use a scalar consequence of semiconvexity. Let F be semiconvex, and suppose that almost everywhere its derivative belongs to a finite collection of continuous functions g_j. Suppose their maximum M is locally Lipschitz and, on a compact interval, each branch either equals M identically or stays below $M - \delta$ for some $\delta > 0$. If $F'_+(s) = M(s)$ at its left endpoint, then $F' = M$ on that interval.

Indeed $G = F - \int_s^u M$ is semiconvex. Its right derivative is right-continuous, has left limits, and obeys $G'_-(u) \le G'_+(u)$. Approaching from the right through the full-measure differentiability set, and taking a constant-index subsequence, gives

$$G'_+(u) \in \{0\} \cup (-\infty, -\delta].$$

At s it is zero. Right-continuity makes the zero state persist to the right. If the negative set had a first infimum after this initial interval, its left limit there would be zero, so its right derivative would be nonnegative and hence zero. Right-continuity would again exclude negative values immediately afterwards, a contradiction. Thus $G'_+ = 0$ throughout, and absolute continuity proves the scalar assertion.

Apply this to $F(u) = S_k(h(s), h(u))$. By Lemma 6.1, its derivative almost everywhere is a k-element subset sum of $v(u)$. The largest such sum is $M_k = \sum_{i \le k} v_i$, and every other subset sum is smaller by at least $v_k - v_{k+1}$. This has a positive minimum on any compact subinterval of I. The forward secant limit gives $F'_+(s) = M_k(s)$. The scalar assertion proves (55). □

Proposition 6.3. *Suppose $v_k > v_{k+1}$ on an open interval containing $[l, r]$. Choose a common orthogonal basis of $h(l), h(r)$, ordered by their decreasing relative weights. Its first k and last $n - k$ vectors span subspaces U, W for which*

$$h(u) = h(u)|_U \oplus h(u)|_W \qquad (l \le u \le r). \tag{56}$$

For every $l \le s < t \le r$, the relative spectrum on U is the first k entries of the full spectrum. The block velocity lists are therefore $(v_1, \dots, v_k)$ and $(v_{k+1}, \dots, v_n)$. Several strict cuts give one simultaneous fixed orthogonal block decomposition, using the same endpoint basis.

Proof. Let τ_U be the wedge of the first k endpoint basis vectors. It realizes the top exterior operator norm between the endpoints. For an intermediate u, exterior norm comparison gives

$$\log \frac{\det(h(u)|_U)(\tau_U)}{\det(h(l)|_U)(\tau_U)} \le S_k(h(l), h(u)),$$
$$\log \frac{\det(h(r)|_U)(\tau_U)}{\det(h(u)|_U)(\tau_U)} \le S_k(h(u), h(r)).$$

The two left sides sum to $S_k(h(l), h(r))$. By (55), the right sides have the same sum. Both inequalities are equalities.

Apply the same argument to the wedge τ_W in the reverse direction. The identity

$$S_{n-k}(h(t), h(s)) = S_k(h(s), h(t)) - d(h(t)) + d(h(s))$$

shows that this reverse partial sum is also saturated along the chain. With the determinant generators from the endpoint basis, we obtain

$$d(h(u)|_U) - d(h(l)|_U) = S_k(h(l), h(u)),$$
$$d(h(u)|_W) - d(h(l)|_W) = d(h(u)) - d(h(l)) - S_k(h(l), h(u)).$$

The direct-sum norm $g_u = h(u)|_U \oplus h(u)|_W$ consequently has the same determinant as $h(u)$, because the decomposition is orthogonal at l. The ultrametric triangle inequality gives $h(u) \le g_u$. Equal determinant and order force equality, proving (56).

Insert the chain l, s, t, r in the exterior norm comparison. The same saturation makes each inequality an equality. Thus the fixed wedge τ_U realizes $S_k(h(s), h(t))$ for every pair. Orthogonality makes the full spectrum the union of the two block spectra, so the U spectrum consists of its top k entries, including multiplicities. Secant limits identify the block velocities. Their squared d_2 distances add to the full squared distance, so the restricted curves are locally Lipschitz. Finally, the same endpoint basis supplies nested prefix and suffix subspaces for several cuts. Their intersections give the asserted simultaneous decomposition. □

6.2. **Second-order scalar approximation.** A good time will mean a time u_0 where all $2n$ global velocity entries are differentiable and where equal entries have equal derivatives, also across the two vertices. These times have full measure: a Lipschitz function has derivative zero almost everywhere on its zero set, and there are only finitely many differences to consider.

Proposition 6.4. *At a good time, group all velocity entries by their value p_g. Let r_g be their common derivative, and set $f_g(s) = p_g s + r_g s^2/2$, where $s = u - u_0$. There are fixed orthogonal subspaces for these groups near u_0, and reference block norms h_{vg}, such that*

$$\sup_{|s|\le h} d_2\big((c_*)_v(u_0+s), Q_v(u_0+s)\big) = o(h^2),$$
$$Q_v(u_0+s) = \bigoplus_g e^{f_g(s)} h_{vg}. \tag{57}$$

In each group the intrinsic velocities are $f_g'(s) + o(h)$, uniformly for $|s| \le h$. The reference pair Q is ordered for small h. Its reduced arrow is a direct sum over the common values p_g and is the actual reduced velocity decomposition at u_0.

Proof. The gaps between the distinct values at u_0 persist nearby. Proposition 6.3 gives fixed orthogonal subspaces for these coarse groups, even when velocities within a group separate at nearby times. Differentiability at u_0 gives, for every entry in group g,

$$|v_{vgi}(u_0+s) - p_g - r_g s| \le \eta(h)|s|, \qquad |s| \le h, \qquad \eta(h) \to 0.$$

Normalize the restricted block curve by $e^{-f_g(s)}$. Its metric speed is exactly the Euclidean norm of the block velocity entries minus $f_g'(s)$, by the spectral secant formula for d_2. The length bound for a locally Lipschitz metric curve therefore bounds its distance from its value at u_0 by $\sqrt{\dim V_{vg}}\eta(h)s^2/2$. Summing over the orthogonal blocks proves (57) and the velocity assertion. The determinant logarithm of each block is C^1, with derivative the sum of its continuous velocity entries.

Choose orthonormal L-bases for the reference block norms. In these bases the original arrow matrix is integral. If a valuation-zero entry connected distinct velocities $p_C \neq p_A$, the arrow inequality for Q would fail linearly on one side of u_0. The $o(s^2)$ norm error in (57) cannot repair such a failure for the actual ordered curve. Thus every nonzero cross-group entry has strictly positive valuation. There are only finitely many entries, so these strict gaps persist for small s. Within a common group the scalar functions f_g at both vertices agree, including their quadratic term. All its arrow inequalities are those at u_0. Hence Q is ordered. Its tangent agrees with that of the actual curve, so its reduced summands are the actual velocity blocks. □

Lemma 6.5. *At a good time, the reference Q in Proposition 6.4 can be repaired at the endpoints of $[u_0-h,u_0+h]$ to an ordered finite piecewise-apartment curve Q_h with the exact endpoints of c_*, such that $Q_h = Q$ on $[u_0-h/2,u_0+h/2]$ and*

$$E(Q_h) - \overline{E}(c_*|_{[u_0-h,u_0+h]}) = o(h^3). \tag{58}$$

Proof. Choose $\delta_h \downarrow 0$ bounding the norm errors in (57) by $\delta_h h^2$ and the block velocity errors by $\delta_h h$. Put $\ell_h = \sqrt{\delta_h}h < h/4$. Keep Q on the central interval and join it to the exact c_* endpoints by canonical bridges of length ℓ_h. The endpoints of each bridge are ordered and have the same fixed block decomposition. The bridge respects that decomposition, since its locus of orthogonal sums is geodesically convex. Each pair of block endpoint norms has a common L-basis, so this is a finite piecewise-apartment comparison curve.

After subtracting f_g, a bridge velocity differs from f_g' by

$$O(\delta_h h^2/\ell_h + \ell_h) = O(\sqrt{\delta_h}h).$$

The first term is the endpoint error divided by the bridge length; the second compares the average quadratic slope to its derivative. The original normalized block speed is $O(\delta_h h)$. Throughout the interval Q_h and c_* are $O(\delta_h h^2)$ apart, including the quadratic chord error $O(\ell_h^2)$ on the bridges. Both curves have a common strict velocity margin for small h.

Use, separately in each block, the identity

$$\ell(z) = \ell(f_g') + \tan(f_g')(z - f_g') + D_\ell(z; f_g').$$

On the strict margin the Bregman remainder is bounded by a constant times $(z-f_g')^2$. Its integral for c_* is $O(\delta_h^2 h^3)$; the bridges of Q_h contribute $O(\delta_h^{3/2}h^3)$, and its central part contributes zero. The linear term is $R\tan(f_g')$ times the derivative of the block determinant difference. Integrate it by parts. Its boundary term vanishes separately in each block, because the repaired endpoints are exact, and its integral is $O(\delta_h h^3)$. The potential difference has the same bound. The original action is its spectral integral by Proposition 4.8; the repaired curve has its ordinary differential action by Lemma 3.3. Combining these estimates proves (58). □

6.3. **Almost-everywhere semistability and straight directions.**

Proposition 6.6. *At almost every time the actual reduced velocity blocks are semistable. A block has type A^m, B^m or C^m, and its position has derivative $-a$, 0 or a, respectively, in the coordinate $t = \sinh u$. In a mixed block the actual reduced arrow is an isomorphism.*

Proof. Work at a good time u_0 and use the quadratic reference and repair above. For a group g, let c_g, a_g be its source and target dimensions, $d_g = c_g + a_g$, and let $R_0 = R(u_0)$. Choose a smooth function η supported in $(-1/2, 1/2)$. On the central part of Q_h, vary all weights of this group by $\varepsilon h^2 \eta(s/h)$. Both signs are admissible for small h: within the group the arrow ratios stay fixed, while the strict cross-group gaps persist. The bridges are unchanged.

Expand the differential action, divide by h^3, and let $h \to 0$ with ε fixed. The linear term is

$$\varepsilon \left\{ -\frac{d_g}{2} \big(R_0 \sec^2 p_g \, r_g + R_0' \tan p_g \big) + q R_0 (c_g - a_g) \right\} \int \eta.$$

For the kinetic term, integrate $R \tan(f_g')$ against the weight derivative; its constant part integrates to zero and its first-order part gives the displayed derivative at u_0. The remainder after division is $O(\varepsilon^2) + o(1)$, uniformly for ε in a fixed bounded interval. Local minimality and (58), followed by $\varepsilon \to 0$ from both sides, imply

$$\frac{d_g}{2} \big(R_0 \sec^2 p_g \, r_g + R_0' \tan p_g \big) = q R_0 (c_g - a_g). \tag{59}$$

Let S be a nonzero subrepresentation of the reduced block, with dimensions c_S, a_S and $d_S = c_S + a_S$. Choose integral basis lifts of S and complements at both vertices. Such changes preserve the scalar reference Q exactly. Reduced entries from S to its complement vanish, so their actual entries are zero or have strictly positive valuation. Entries in the other direction become more contracting when the weights on S decrease. Thus, for $\eta \geq 0$ and $\varepsilon \geq 0$, the following variation is ordered for small h: add $(d_S/d_g)\varepsilon h^2 \eta(s/h)$ to the whole group and subtract $\varepsilon h^2 \eta(s/h)$ on S.

All unperturbed group velocities are f_g', and the total weight shift, with dimension multiplicities, is zero. Its kinetic first variation therefore vanishes pointwise. The potential coefficient after division by h^3 is

$$\varepsilon q R_0 \left\{ \frac{d_S}{d_g} (c_g - a_g) - (c_S - a_S) \right\} \int \eta.$$

Again the remainder is $O(\varepsilon^2) + o(1)$. If the bracket were negative, choose $\varepsilon > 0$ small enough that its linear term dominates, then let $h \to 0$. This contradicts local minimality and (58). Hence

$$\frac{c_S - a_S}{c_S + a_S} \leq \frac{c_g - a_g}{c_g + a_g} \quad \text{for every nonzero subrepresentation } S. \tag{60}$$

For the charges in (2), this is precisely the semistability inequality: the phase increases strictly with the ratio $(c - a)/(c + a)$.

Every finite-dimensional A_2 representation is the direct sum of its kernel copies of C, image copies of B, and cokernel copies of A, by choosing complements to the kernel and image. Since their phases are distinct, semistability permits only one of these types. Thus a shared velocity block is actually B^m, with invertible reduced arrow, rather than only equal dimensions. Finally, the derivative in t of $x = R \tan v$ is, at the good time,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{R_0' \tan p_g + R_0 \sec^2 p_g \, r_g}{R_0} = a \frac{c_g - a_g}{c_g + a_g}.$$

This gives $-a, 0, a$ for the three types. Good times have full measure, which proves the assertion. □

6.4. **Finiteness of the support.**

Proposition 6.7. *The spectral positions of c_* form a finite balanced weighted straight graph. It has at most $2n^2$ distinct finite event times and at most n^2 essential geometric vertices.*

Proof. Write $A_i(t)$ and $C_j(t)$ for the two ordered position lists. They are locally Lipschitz. Proposition 6.6 gives, on one common set of full measure,

$$A_i' \in \{-a, 0\}, \qquad C_j' \in \{0, a\}.$$

An A position absent from the C list has derivative $-a$, and an unshared C position has derivative a. At a shared position the two multiplicities agree and the derivative is zero.

Each difference $C_j - A_i$ is nondecreasing. Its zero set is an interval, possibly empty, a point, a ray or the whole line. Moreover, if $s < t$ lie in that zero set, then

$$A_i(s) \geq A_i(t) = C_j(t) \geq C_j(s) = A_i(s),$$

so both positions are constant between them. Let D be the set of finite endpoints of these n^2 zero sets. Then $|D| \leq 2n^2$. On a component of $\mathbb{R} \setminus D$, each opposite-type pair either coincides throughout or never coincides. A paired branch is constant; an unpaired A branch has derivative $-a$ almost everywhere and hence is exactly affine, and an unpaired C branch is similarly affine of slope a.

These pieces define the graph with coincident positions combined and their multiplicities retained. A constant branch is counted once as a B edge, with its common multiplicity. Parallel branches of one free type do not intersect; a free branch cannot meet a constant branch in the interior of a component, since that would be an opposite-type contact. There are therefore no additional crossings to insert there. The known tails supply its exterior rays. This proves finiteness before any appeal to honeycomb combinatorics.

At a vertex isolate its position from the other finitely many positions at that time. Continuity implies that the same A indices and the same C indices enter and leave this neighborhood. Writing m_X^- and m_X^+ for the incoming and outgoing multiplicities of type X, we obtain

$$m_A^- + m_B^- = m_A^+ + m_B^+, \qquad m_C^- + m_B^- = m_C^+ + m_B^+.$$

The upward direction vectors

$$z_A = (-a/2, 1/2), \qquad z_B = (0, 1), \qquad z_C = (a/2, 1/2)$$

satisfy $z_B = z_A + z_C$. The preceding identities give their weighted balance. There are no finite univalent vertices, and the tails have degree n.

We may now apply the finite honeycomb construction of [KT99, §3.1, Lemmas 4 and 6–8, and Theorem 1]. Its dual is a subdivision of the lattice triangle of side n. The two-dimensional cells correspond to essential geometric vertices, with each cell a union of elementary triangles. The entire triangle has n^2 elementary triangles. Every cell contains at least one, proving the bound n^2. Degenerate or crossing vertices correspond to larger cells; artificial bivalent subdivisions are removed. This last count bounds vertices, whereas the earlier direct estimate bounds only their distinct time coordinates. □

6.5. **From finite support to finite apartments.**

Proposition 6.8. *The Hahn minimum has a finite fixed-apartment presentation. On each closed finite piece its weights extend real analytically beyond the endpoints. On every open piece its actual reduced velocity blocks are semistable and have the straight directions prescribed by their charges.*

Proof. Choose finite cutoffs in the known algebraic tails and subdivide between them at the support event times. On an open piece (b, c), the ordered position branches are affine in t. Distinct branches do not intersect there, and coincident branches agree throughout. Thus the velocity groups have constant multiplicities and functions

$$p_g(u) = \arctan(\alpha_g \tanh u + \beta_g \operatorname{sech} u), \qquad \alpha_g \in \{-a, 0, a\}. \tag{61}$$

Each extends analytically beyond the closed piece. The gaps between distinct groups are positive in its interior but may vanish at either endpoint.

For each cut k between distinct groups, (55) holds for all $b < s < t < c$. Continuity of the norms, spectra and velocities extends it to $b \le s < t \le c$. Choose one common endpoint basis, ordered by relative weights. The extremal-wedge proof of Proposition 6.3 uses only these saturated identities on closed chains. It therefore gives one simultaneous orthogonal block decomposition on the whole closed interval, even when a velocity gap vanishes at an endpoint. No lower bound for that endpoint gap is needed. If there is only one group, take the entire space as its block.

In block g all velocities equal p_g. Fix an interior time u_*, set $f_g(u) = \int_{u_*}^{u} p_g(s)\,\mathrm{d}s$, and let h_g be the block norm at u_*. The normalized block curve $e^{-f_g(u)} c_*(u)|_{V_g}$ has metric speed zero. Its local Lipschitz continuity and the metric length bound make it constant. Hence

$$c_*(u)|_{V_g} = e^{f_g(u)} h_g$$

throughout the interior and, by continuity, at both endpoints. Choose an orthogonal basis for each split norm h_g. Their union is one fixed apartment on the entire closed piece, and its weights are f_g plus constant offsets. This proves the claimed analytic extension without taking a limit of bases on an exhaustion.

In these fixed bases, equal-velocity coordinates have constant relative weights, so their normalized reduced arrow matrix is constant. A nonzero entry between distinct groups has strictly positive norm gap at every interior time: an interior zero of its nonnegative smooth gap would have zero derivative and would force the two velocities to agree. Thus these fixed groups are the actual reduced blocks everywhere on the open piece. Their almost-everywhere semistability from Proposition 6.6, together with the constant reduced matrices, proves semistability throughout. Adjoin the finitely many tail apartments to complete the presentation. □

6.6. **Finite descent and identification of the original minimum.** The preceding presentation is finite, but its matrices lie in L. The following finite realization preserves more than spectra: it retains the actual reduced arrows and the full endpoint norms.

Lemma 6.9. *Let an ordered continuous L-norm pair on a compact interval have finitely many fixed-apartment pieces, with prescribed real C^1 weights and strict velocity margins. Suppose its endpoint norms are extensions of specified split K-norms. There is a continuous ordered K-norm pair with the same partition and weights, exactly those full endpoints, the same determinant functions, the same normalized reduced arrow matrices on all pieces, and*

the same normalized reduced transition matrices at their joins. In particular its differential action is unchanged.

Proof. Use the same ambient K-frame at both quiver vertices. Let P_{vj} be the finitely many original apartment basis matrices. Choose orthonormal K-frames $F_{v,-}, F_{v,+}$ for the prescribed outer endpoint norms. Introduce matrix variables Q_{vj} and R_{vj} with $Q_{vj}R_{vj} = R_{vj}Q_{vj} = I$. Consider the finite list consisting of the entries of

$$R_{Aj}Q_{Cj}, \qquad R_{vj}Q_{v,j+1}, \quad R_{v,j+1}Q_{vj},$$
$$R_{v,\mathrm{first}}F_{v,-}, \quad F_{v,-}^{-1}Q_{v,\mathrm{first}}, \quad R_{v,\mathrm{last}}F_{v,+}, \quad F_{v,+}^{-1}Q_{v,\mathrm{last}},$$

and of the determinants $\det Q_{vj}$. Evaluate them at the original matrices P_{vj} and their inverses. Preserve every zero by an exact equation. For a nonzero value x, put $\rho = \nu(x)$ and $a_x = \mathrm{res}(T^{-\rho}x)$, and impose the valued-field condition

$$T^{-\rho}x - a_x \in \mathfrak{m}. \tag{62}$$

All parameters lie in K, including the finitely many real monomials and complex residues. The original L-matrices satisfy this finite formula, so elementarity gives invertible K-matrices satisfying it. It preserves each entry valuation and leading residue, and each determinant valuation.

For a matrix M from a source basis of weights w_j to a target basis of weights z_i, nonexpansiveness is the finite collection of inequalities

$$\nu(M_{ij}) + w_j - z_i \geq 0 \tag{63}$$

for its nonzero entries. The valuations and weight functions are unchanged. Thus the transferred arrow is nonexpansive at every time. Both directions of each transition were included, so their inequalities prove equality of the two norm descriptions at each join. The same two-way argument with the fixed endpoint frames proves equality with the full prescribed endpoint norms, on every vector. It is stronger than preservation of their spectra.

A normalized arrow entry is $T^{w_j - z_i}M_{ij}$. If its valuation is positive, its reduction is zero; if it is zero, its reduction is the leading residue preserved in (62). Hence all the reduced matrices agree in the indexed bases, both on pieces and at joins. Equation (13) shows that the determinant functions are unchanged. The weights and their derivatives are also unchanged, so the differential actions agree. No time variable or infinite family of norm inequalities was quantified: the finite entry data imply them for the fixed weight functions. □

Theorem 6.10. *For every full split boundary triple, the original K-minimum c_* has a finite regular harmonic presentation as in Definition 1.1. It has the finite weighted support of Proposition 6.7, satisfies the normalized ends, and obeys the stronger $O(h^2)$ midpoint estimate at every join.*

Proof. Choose $l < 0 < r$ inside the established fixed K-apartment tails. Proposition 6.8 gives a finite Hahn presentation on $[l, r]$. Apply Lemma 6.9 to its exact split K endpoint norms, and append the actual known K tails outside this interval. Call the resulting curve z_K. It is ordered and has the same weights, determinant functions and reduced labels as the Hahn minimum in the middle. Its tails are unchanged. We show that it is the actual K-minimum, rather than merely a curve with the same numerical data.

For either finite presentation, its differential J is finite. On the negative tail the gap is zero and the velocities are $O(e^u)$, so the integrand is $O(e^u)$. On the positive tail the velocity difference from $\epsilon_v\beta$ is $O(e^{-u})$, so the Bregman integrand is $O(e^{-u})$ as well. The norm errors from the reference tails have the same rates.

Cut the actual tails at $\pm V$, keep the middle curve, and join to the exact reference tails on intervals of length one. At the negative end the common norms can be bridged together, so their gap stays zero and their velocities are $O(e^{-V})$. At the positive end both endpoint pairs are ordered; each vertex bridge is in its fixed tail apartment. Relative to the reference velocity its error is $O(e^{-V})$, since the endpoint norm errors have that size and $\beta' = O(e^{-2V})$. The bridge and removed tail costs are $O(e^{-V})$. These curves belong to $\mathcal{Q}$ and converge uniformly to the given finite presentation. In particular,

$$J_*^L \le J_{\mathrm{diff}}(c_*^L).$$

For the reverse inequality, take a global polygonal minimizing sequence converging uniformly to c_*^L. On each fixed $[-V, V]$, the partition lower bound gives

$$E_{\mathrm{diff},[-V,V]}(c_*^L) \le \liminf_m E_{[-V,V]}(p_m).$$

The endpoint determinant terms converge. Subtracting the finite correction in (35) gives the same lower bound for truncated J. Since its integrands are nonnegative, $J_{\mathrm{diff},[-V,V]}(c_*^L) \le J_*^L$. Monotone convergence as $V \to \infty$ proves $J_{\mathrm{diff}}(c_*^L) = J_*^L$.

Scalar extension equates the two field infima. The transferred compact actions and the identical tails now give

$$J_{\mathrm{diff}}(z_K) = J_{\mathrm{diff}}(c_*^L) = J_*^K.$$

The same exact-tail recovery over K places z_K in $\mathcal{Y}_K$ and gives $\overline{J}(z_K) \le J_*^K$. Uniqueness in Theorem 3.4 forces $z_K = c_*^K$ as actual norm curves. Thus the original minimum has the finite K presentation and actual semistable labels asserted here. Its midpoint estimate is Proposition 4.6; it does not depend on the transferred coordinates. The already proved tails give precisely the full normalized end limits. □

The construction has supplied algebraic edge labels on a finite support. To finish the path correspondence in Theorem 1.2, we must construct the local vertex filtrations and prove that every finite regular path with first-order matching is this minimum. Those are distinct uses of the matching condition, addressed next.

7. Vertex filtrations and uniqueness

We first construct the local residue data at the vertices. We then prove a converse: a finite regular harmonic path satisfying first-order matching is the variational minimum. Both arguments use the full norm motion, rather than only its unordered velocity spectrum.

7.1. The residue representation at a vertex.

Proposition 7.1. *Let y be a finite regular harmonic path satisfying (10). At every support vertex it determines an actual reduced A_2 representation with two increasing-phase filtrations. Their quotients are the incident incoming and outgoing edge objects, with their actual arrows. These identifications extend along edges and remove artificial subdivisions. Changes of apartment bases give isomorphic data.*

Proof. Fix a subdivision time u_0, put $s = u - u_0$, and choose $y_v(u_0)$-orthonormal reference bases at the two vertices. Normalize the left and right apartment frames in the same way. Every transition matrix P lies in $\mathrm{GL}_n(\mathcal{O})$, with invertible reduction $\overline{P} \in \mathrm{GL}_n(\mathbb{C})$. The matrix $\overline{P}^{-1}P$ is $I + N$, where all nonzero entries of N have positive valuation. For the finitely many matrices and inverses choose a common positive lower bound δ for those valuations. As in Lemma 5.2, replacing P by $\overline{P}$ preserves exactly every weighted norm whose weight spread

is less than δ. Our one-sided weights tend to zero. Thus, on a small neighborhood, the two given norm germs are exactly norms defined by weighted filtrations of the reference residue spaces $V_v = \mathbb{C}^n$.

Concretely, such a filtration norm is diagonal with real weights in a complex basis, extended to the valued space using the reference frame. Two finite filtrations admit a simultaneously adapted complex basis, obtained by splitting their intersections. Their canonical norm geodesic therefore interpolates the weights in that basis. Multiplication of all weights by a positive number preserves this locus and scales its metric by that number. This description concerns the finitely many given germs, not a neighborhood of all norms.

Write their first-order expansions as

$$w^+_{vi}(s) = p^+_{vi}s + O(s^2), \qquad w^-_{vi}(s) = p^-_{vi}s + O(s^2).$$

Scale weights by $1/h$ at $s = -h, h$. The endpoint norms converge to coefficient norms k^-_v, k^+_v with weights $-p^-_{vi}$ and p^+_{vi}. Matching and continuity of the midpoint give

$$\operatorname{mid}(k^-_v, k^+_v) = 1, \tag{64}$$

where 1 denotes the norm with zero weights in the reference frame. In a simultaneously adapted complex basis, the paired weights therefore sum to zero. The two first-derivative lists agree with multiplicities, and their filtrations are opposite. In particular, if

$$F^+_v(\le p) = \operatorname{span}\{\text{positive-side first weights } \le p\},$$
$$F^-_v(\ge p) = \operatorname{span}\{\text{negative-side first weights } \ge p\},$$

then their intersections give the canonical decomposition

$$V_v = \bigoplus_p V_{v,p}, \qquad V_{v,p} = F^+_v(\le p) \cap F^-_v(\ge p). \tag{65}$$

Here the first flag is the sum of blocks of index at most p, and the second is the sum of blocks of index at least p.

Let $f_0 : V_C \to V_A$ be the reduction of the original arrow in the reference frames. For $s > 0$, nonexpansiveness forces f_0 to preserve the increasing flag F^+; for $s < 0$ it forces preservation of the decreasing flag F^-. Indeed a nonzero reduced entry must satisfy $p_A \le p_C$ on the positive side and $p_A \ge p_C$ on the negative side. Hence f_0 preserves every summand in (65). The object

$$E_p = (V_{C,p} \xrightarrow{f_0} V_{A,p}) \tag{66}$$

is an actual subrepresentation of the reduction. Its support position is $(x_0, t_0) = (R(u_0)\tan p, \sinh u_0)$.

Fix this p and one side of the vertex. Its finer apartment frame induces a frame of the p-graded quotient, identified with E_p by (65). A branch of slope $\alpha \in \{-a, 0, a\}$ through (x_0, t_0) has

$$w'_\alpha(u) = \arctan \frac{x_0 + \alpha(\sinh u - \sinh u_0)}{R(u)}.$$

Its second derivative at the vertex is

$$r_\alpha = (\alpha - \tanh u_0 \tan p)\cos^2 p. \tag{67}$$

This increases strictly with α, so its order is A, B, C on both sides of the vertex. The relevant expansions are $ps + r_\alpha s^2/2 + O(s^3)$; the square has the same sign on both sides.

Inside this p block, a nonzero reduced arrow entry must obey $r_{\text{target}} \le r_{\text{source}}$, or its norm inequality would fail for every small nonzero s on that side. Therefore the spans of blocks

with $r \le r_\alpha$ form a filtration by subrepresentations of E_p. Its successive quotients have the actual arrow reductions of the incident edges. To see the latter assertion, coordinates with the same p and α have exactly the same derivative function and, after normalization at u_0, the same weight function. Entries inside that block retain their reduction. Allowed entries between distinct second weights are strictly contracting off the vertex; entries with positive initial valuation remain strictly contracting on a sufficiently small neighborhood. The same observation handles distinct first weights. Thus each graded quotient is the actual semistable edge object, not merely an object with the same dimension vector. Omitting zero factors gives increasing phases $1/6, 1/2, 5/6$.

Doing this on the two sides gives two filtrations of the same E_p. On each open piece the normalized residue arrow is constant, giving identifications along an edge. At an artificial subdivision with one continuing branch, each filtration has just the one factor E_p, identifying the adjacent labels. Coincident branches are treated with their combined multiplicity. Finally, the first flags are intrinsic one-sided norm germs; the second flags are their graded limits after subtracting ps and scaling weights by $2/s^2$. The intersection decomposition and these limits show independence of presentation, up to isomorphism of the objects, filtrations and edge identifications. □

These are the local residue-category data described in [HKS24, §4.4]. The proposition does not identify them with a specified global Novikov-field Fukaya object or a Maurer–Cartan solution. Its input is the norm matching condition; it does not prove that abstract vertex filtrations imply that condition.

7.2. **Local variational comparison on open pieces.**

Lemma 7.2. *A finite regular harmonic path is locally the exact-endpoint relaxed minimum away from finitely many times on each compact interval. First-order matching is not needed for this assertion.*

Proof. Its straight branches make the one-sided weight functions analytic beyond their finite endpoints. Apply the analytic common-basis assertion of Lemma 2.5 to the two vertex norms on each piece. Refine further at the zeros of the analytic coordinate gaps $g_i = w_{Ci} - w_{Ai}$. Each gap on a remaining open piece is either identically zero or strictly positive. Both vertices now use the same ambient basis. Put $P_{vi} = R \tan w'_{vi}$.

A zero gap gives a reduced identity arrow on that coordinate. Its semistable block must have type B, so

$$w_{Ai} = w_{Ci}, \quad P_{Ai} = P_{Ci}, \quad P'_{Ai} = P'_{Ci} = 0.$$

At a positive gap the reduced source and target are C and A summands. Semistability of their full velocity blocks therefore forces

$$P'_{Ai} = -aR, \qquad P'_{Ci} = aR.$$

These statements use the actual coordinate arrows in the common basis.

On a compact subinterval $[l, r]$ of this piece, first compare with an ordered curve z in the same apartment and with the same endpoints. Write $\delta_{vi} = z_{vi} - w_{vi}$. The Bregman identity and integration by parts give

$$\begin{aligned} E(z) - E(y) = \frac{1}{2}\sum_{v,i} [P_{vi}\delta_{vi}]_l^r + \frac{1}{2}\int_l^r R \sum_{v,i} D_\ell(z'_{vi}; w'_{vi})\,\mathrm{d}u \\ + q \sum_{i: g_i \equiv 0} \int_l^r R(z_{Ci} - z_{Ai})\,\mathrm{d}u. \end{aligned} \tag{68}$$

There is no basis-volume offset in the gap because both vertices use the same frame. On positive-gap coordinates the momentum derivatives cancel the potential; on zero-gap coordinates they leave the last displayed term. The endpoint term vanishes, and the other terms are nonnegative. This proves the comparison for finite action; the infinite case is immediate. A strict margin is needed only for the reference curve in the linear term.

For an arbitrary ordered comparison polygon, project its nodes onto this common apartment at both vertices by the line-sign projection. It preserves order, endpoints and determinant gap, and decreases the spectral kinetic action. Apply (68) to the resulting polygon. Passing to all-norm recovery sequences and using regular-curve recovery proves that y is the local relaxed minimum. □

7.3. **The sign of the boundary terms.** We record the needed CAT(0) fact with its normalization. For an apartment ray $\gamma(s)$ from o with logarithmic velocity $d = (d_i)$, put $D = \|d\|_2$. Its speed-scaled Busemann function is

$$B_\gamma(k) = D \lim_{s\to\infty} \big(d_2(k, \gamma(s)) - sD\big).$$

For $D = 0$ set this function to zero. The limit exists by the triangle inequality and is finite. If k belongs to the same apartment, with weight difference δ from o, the Euclidean distance formula gives

$$B_\gamma(k) = -\sum_i d_i\delta_i. \tag{69}$$

Lemma 7.3. *Suppose two apartment rays γ_-, γ_+ from o have opposite weighted germs and the same speed D. Then*

$$B_{\gamma_-}(k) + B_{\gamma_+}(k) \geq 0 \tag{70}$$

for every norm k. At a matched join of a finite regular path, the outward momentum germs $-R(u_0)\tan p^-$ and $R(u_0)\tan p^+$ satisfy this hypothesis, including repeated velocities and the scalar determinant direction.

Proof. Opposite germs form a local geodesic through o. The concatenation of the two rays is a local geodesic everywhere, and hence a geodesic line in the CAT(0) norm space. Thus $d_2(\gamma_-(s), \gamma_+(s)) = 2sD$. The triangle inequality through k, followed by the defining limit, proves (70).

For the final assertion, the proof of Proposition 7.1 identifies opposite first-derivative flags in one common residue frame. The odd strictly increasing function $p \mapsto R(u_0)\tan p$ preserves their order, repetitions and opposite weights. Their momentum rays therefore have opposite germs and equal Euclidean speed. The finite residue-matrix argument used there makes this an exact local geodesic assertion for the given apartment rays. No strict Weyl chamber or trace-free normalization is required. □

7.4. **The converse across all joins.**

Theorem 7.4. *A finite regular harmonic path satisfying first-order matching is the unique exact-endpoint relaxed minimum on every compact interval. Competitors may change apartments and may take values in the complete all-norm space.*

Proof. Let y be the path on $[a, b]$. First let z be an ordered split polygon of finite action with the same full endpoints. Refine the pieces of y and z at the finitely many exceptions in Lemma 7.2. At each vertex separately, apply the analytic common-basis lemma to y_v and z_v on every closed piece. The weights of y are analytic past the endpoints, and those of z are

affine in its original frame. A further finite subdivision therefore gives common fixed bases for y_v, z_v, valid at the endpoints as well as in the interior. The two quiver vertices need not use the same basis.

Write their weights in these bases as w_{vi}, z_{vi} and set

$$\delta_{vi} = z_{vi} - w_{vi}, \qquad P_{vi} = R \tan w'_{vi}.$$

For $0 \le s \le 1$, canonical pointwise interpolation y_s between y and z has weights $w_{vi} + s\delta_{vi}$ in these bases. It is ordered by convexity of the norm order. It is a finite regular comparison curve, with

$$E(y_s) \le (1-s)E(y) + sE(z).$$

The finitely many compact velocity margins justify differentiating at $s = 0$:

$$\left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{0+} E(y_s) = \int_a^b L(u)\,\mathrm{d}u, \qquad L = \frac{1}{2}\sum_{v,i} P_{vi}\delta'_{vi} + qR\sum_i(\delta_{Ci} - \delta_{Ai}). \tag{71}$$

On an open cell I put $H = \frac{1}{2}\sum_{v,i} P_{vi}\delta_{vi}$. Take a nonnegative smooth ϕ compactly supported in I, and replace the constant interpolation parameter by $\varepsilon\phi(u)$. For small $\varepsilon \ge 0$ this stays in $[0,1]$, preserves order, and has a strict velocity margin. It equals y outside a compact subset of I. Local minimality from Lemma 7.2 gives

$$0 \le \int_I (\phi L + \phi' H)\,\mathrm{d}u = \int_I \phi(L - H')\,\mathrm{d}u.$$

The integrand is continuous, so $L - H' \ge 0$ on the cell. This uses an admissible variation of the pair and does not require the velocity orders at the two vertices to agree.

At an interior subdivision time u_0, let $o_v = y_v(u_0)$ and $k_v = z_v(u_0)$. In the two respective common endpoint apartments choose rays from o_v with outward momentum velocities

$$-P_v(u_0-), \qquad P_v(u_0+).$$

They have opposite germs by Lemma 7.3, also at artificial refinements inside an original piece. Both o_v and k_v belong to each corresponding apartment. Formula (69) consequently gives the exact identities

$$B_{v,-}(k_v) = \sum_i P_{vi}(u_0-)\delta_{vi}(u_0-), \qquad B_{v,+}(k_v) = -\sum_i P_{vi}(u_0+)\delta_{vi}(u_0+).$$

The left and right bases can differ. The Busemann inequality, not equality of the coordinate expressions, gives

$$H(u_0-) - H(u_0+) = \frac{1}{2}\sum_v \{B_{v,-}(k_v) + B_{v,+}(k_v)\} \ge 0. \tag{72}$$

Integrate $L \ge H'$ over every cell and sum. The outer boundary terms vanish by equality of the full endpoint norms, and (72) makes all internal contributions nonnegative. Thus $\int_a^b L \ge 0$. Convexity in s and (71) imply

$$E(z) - E(y) \ge \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{0+} E(y_s) \ge 0.$$

This proves comparison with every split polygon.

For a finite-action polygon with arbitrary all-norm nodes, replace it by the minimum on its fixed partition. Its endpoints are the split norms of y, so Proposition 3.1 supplies a split polygon of no larger action with those exact endpoints. The preceding comparison applies. This replacement need not be close to the original polygon; only its action and endpoints

are used. Every polygon in an exact-endpoint recovery sequence therefore has action at least $E(y)$. Regular-curve recovery shows that y attains the relaxed infimum. Strict midpoint stability then gives uniqueness in the complete relaxed domain. □

Theorem 7.5. *Every finite regular harmonic path with first-order matching and the normalized ends* (8) *equals c_* as an actual full norm path.*

Proof. Let y be such a path. Its finite apartment partition gives one fixed apartment at each vertex on each distant tail. A coordinate branch there has

$$w'(u) = \arctan(\alpha \tanh u + b \operatorname{sech} u), \qquad \alpha \in \{-a, 0, a\}. \tag{73}$$

At the negative end the coordinate weights have finite limits. If $\alpha \neq 0$, the derivative would tend to the nonzero constant $\arctan(-\alpha)$, contradicting those limits. Thus all incoming branches have $\alpha = 0$ and type B. The reduced arrow has invertible graded blocks, hence is itself invertible. For two ordered split norms, invertibility of the reduced identity arrow means that all relative gaps vanish. Consequently $y_A = y_C$ on this tail. Equation (73) and the full incoming limit give

$$d_2(y_v(u), h_B) = O(e^u), \qquad w'_{vi}(u) = O(e^u).$$

One common apartment can be used for the equal pair.

On the positive tail, $w_{vi} - \epsilon_v F$ has a finite limit. Its derivative tends to $\arctan \alpha - \epsilon_v \pi/3$, so $\alpha = \epsilon_v a$. The limiting reference norm belongs to the same apartment, by closedness after scalar normalization. Integrating the derivative difference gives

$$d_2(y_v(u), r_v(u)) = O(e^{-u}), \qquad w'_{vi}(u) - \epsilon_v \beta(u) = O(e^{-u}).$$

These rates show that the nonnegative differential action $J_{\mathrm{diff}}(y)$ is finite and its tails outside $[-U, U]$ contribute $O(e^{-U})$.

The exact-reference recovery in the proof of Theorem 6.10 applies verbatim: keep y on $[-U, U]$, bridge to the reference tails over intervals of length one, and use the references beyond them. The resulting $y_U \in \mathcal{Q}$ converge uniformly to y, and $J(y_U) \to J_{\mathrm{diff}}(y)$. In particular $y \in \mathcal{Y}$ and

$$J_* \leq \overline{J}(y) \leq J_{\mathrm{diff}}(y). \tag{74}$$

Fix any $z \in \mathcal{Q}$ and choose V beyond its cutoff and the tail cutoffs of y. Construct z_V by keeping z on $[-V, V]$, using y outside $[-V-1, V+1]$, and bridging

$$y(-V-1) \longrightarrow (h_B, h_B), \qquad r(V) \longrightarrow y(V+1)$$

on the two intervening unit intervals. These are ordered product geodesics in the respective tail apartments. The same tail estimates, together with $\beta' = O(e^{-2V})$, give

$$J_{\mathrm{diff}}(z_V) \leq J(z) + C_y e^{-V}. \tag{75}$$

The constant is independent of z once V exceeds its cutoff.

On $[-V-1, V+1]$ the two curves y, z_V have exactly the same endpoint norms. Theorem 7.4, with regular-curve recovery for z_V, gives $E(y) \leq E(z_V)$ on this compact interval. Convert this comparison to truncated J using (35); the endpoint determinant gaps agree, so their finite corrections cancel. Outside the interval the two curves are identical. Adding their finite common tails yields

$$J_{\mathrm{diff}}(y) \leq J_{\mathrm{diff}}(z_V) \leq J(z) + C_y e^{-V}.$$

Let $V \to \infty$ and take the infimum over $z \in \mathcal{Q}$. Together with (74), this proves $\overline{J}(y) = J_{\mathrm{diff}}(y) = J_*$. Uniqueness of the whole-line relaxed minimum gives $y = c_*$. The argument

compares finite renormalized quantities and never subtracts divergent whole-line unrenormalized actions. □

Proof of Theorem 1.2. Theorem 6.10 constructs the finite regular harmonic path, with its full normalized ends and quadratic midpoint estimate. Proposition 6.7 gives its balanced support and vertex bound; Proposition 7.1 supplies the actual vertex objects, filtrations and edge identifications. The tail statements are Propositions 5.1 and 5.6. Theorem 7.5 proves uniqueness in the stated class. Analytic weights follow from Proposition 6.8 and the finite transfer, which preserves the weight functions.

Finally, simultaneous $\mathrm{GL}_n(K)$ pullback preserves order, relative spectra, determinant gaps, the reference curves and the relaxed action. It carries $\mathcal{Q}$ and $\mathcal{Y}$ for one triple onto those for the transformed triple. Uniqueness therefore makes the constructed path equivariant. This is equivariance of full norms with their given framing, without passing to a quotient by change of basis. □

8. Examples and a collapsing bounded face

The following examples serve different purposes. Scalar sums give explicit norm paths and a visible change of support topology. The subsequent triples lie outside that simultaneous-apartment setting; their paths are supplied by the theorem, without an asserted explicit formula for their supports.

8.1. **The scalar junction map.**

In the original coordinate put

$$\lambda(x,t) = \frac{1}{\sqrt{1+t^2}} \arctan \frac{x}{\sqrt{1+t^2}}, \qquad A_\mu(t) = \int_0^t \lambda(\mu s, s)\,\mathrm{d}s.$$

Thus $A_{\pm a}(t) = \pm F(\operatorname{arsinh} t)$ for $t \geq 0$. A scalar Y meeting at (p,τ) has incoming position p and outgoing positions $p - a(t-\tau)$ and $p + a(t-\tau)$. Given incoming logarithmic norm b_0, set $k = b_0 + \int_{-\infty}^{\tau} \lambda(p,s)\,\mathrm{d}s$. Its common incoming weight is $b_0 + \int_{-\infty}^{t} \lambda(p,s)\,\mathrm{d}s$ up to τ, and its outgoing weights are

$$w_v(t) = k + \int_\tau^t \lambda(p + \epsilon_v a(s-\tau), s)\,\mathrm{d}s.$$

These weights agree to first order at τ, and their gap is positive for $t > \tau$. Their normalized outgoing constants are $b_0 + d_A$ and $b_0 + d_C$, where

$$d_v = \int_{-\infty}^{\tau} \lambda(p,s)\,\mathrm{d}s + \int_\tau^\infty \big[\lambda(p + \epsilon_v a(s-\tau), s) - \lambda(\epsilon_v a s, s)\big]\,\mathrm{d}s - A_{\epsilon_v a}(\tau). \tag{76}$$

The integrands at the ends are $O(|s|^{-2})$.

Proposition 8.1. *The map $(p,\tau) \mapsto (d_A, d_C)$ is a smooth global diffeomorphism of $\mathbb{R}^2$. Thus each scalar full boundary triple has one scalar Y, which is the path of Theorem 1.2.*

Proof. Write $f(x,s) = (1 + x^2 + s^2)^{-1} = \partial_x \lambda(x,s)$, and set

$$P = \int_{-\infty}^{\tau} f(p,s)\,\mathrm{d}s, \quad L = \int_\tau^\infty f(p - a(s-\tau), s)\,\mathrm{d}s, \quad R_+ = \int_\tau^\infty f(p + a(s-\tau), s)\,\mathrm{d}s.$$

Differentiating the convergent integrals gives

$$d_{A,p} = P + L, \quad d_{A,\tau} = aL, \qquad d_{C,p} = P + R_+, \quad d_{C,\tau} = -aR_+.$$

Their Jacobian is $-a\{P(L+R_+)+2LR_+\}<0$. For a global argument use $y=a\tau$ and output coordinates $U=d_A+d_C$, $V=d_A-d_C$. The derivative matrix is

$$H=\begin{pmatrix}2P+L+R_+ & L-R_+\\ L-R_+ & L+R_+\end{pmatrix},\qquad z^THz=2Pz_1^2+L(z_1+z_2)^2+R_+(z_1-z_2)^2.$$

It is symmetric positive definite. Hence $(U,V)=\nabla G$ for a smooth strictly convex function G on $\mathbb{R}^2$. Put $\rho=\sqrt{1+p^2+\tau^2}$. In each integral retain a segment of length ρ adjacent to τ. On those segments $1+x^2+s^2\le C\rho^2$, with a constant depending only on a. Thus $P,L,R_+\ge c/\rho$, and

$$H\ge\frac{c'}{\sqrt{1+p^2+y^2}}I.$$

At the origin $(U,V)=0$. Along every unit ray this implies $e\cdot\nabla G(re)\ge c'\operatorname{arsinh} r$. Integration makes $G(q)-w\cdot q$ coercive for every $w\in\mathbb{R}^2$. Its unique minimum solves $\nabla G(q)=w$. The inverse function theorem gives a smooth inverse everywhere. The scalar Y is a finite regular matched path with the prescribed ends, so Theorem 7.5 identifies it with the constructed minimum. □

8.2. **An explicit degree-three bounded face.** Fix $p\ge0$ and $(r_1,r_2,r_3)=(-p,0,p)$. In the standard basis of K^3 define

$$w_{vi}(t;p)=\begin{cases}\displaystyle\int_{-\infty}^{t}\lambda(r_i,s)\,\mathrm{d}s, & t\le0,\\ \displaystyle\int_{-\infty}^{0}\lambda(r_i,s)\,\mathrm{d}s+\int_0^t\lambda(r_i+\epsilon_v as,s)\,\mathrm{d}s, & t\ge0.\end{cases}\tag{77}$$

Let $h_v(t;p)$ be the corresponding diagonal norms. Their full incoming norm is the standard maximum norm h_0. Their outgoing normalized norms are diagonal in that same basis, with weights

$$b_{vi}(p)=\int_{-\infty}^{0}\lambda(r_i,s)\,\mathrm{d}s+\int_0^\infty\big[\lambda(r_i+\epsilon_v as,s)-\lambda(\epsilon_v as,s)\big]\,\mathrm{d}s.\tag{78}$$

These formulas specify the norms on every vector, including their scales. Each coordinate is the scalar Y at $(r_i,0)$. The pair is equal for $t\le0$ and ordered for $t\ge0$, since

$$w_{Ci}(t;p)-w_{Ai}(t;p)=\int_0^t[\lambda(r_i+as,s)-\lambda(r_i-as,s)]\,\mathrm{d}s>0\quad(t>0).$$

It has first-order matching at zero and smooth weights elsewhere. Theorem 7.5 identifies it with the unique matched path for (78).

For $p>0$ the Y vertices and crossing vertices are

$$\begin{gathered}Y_1=(-p,0),\quad Y_2=(0,0),\quad Y_3=(p,0),\\ P_{12}=(-p/2,p/(2a)),\quad P_{23}=(p/2,p/(2a)),\quad P_{13}=(0,p/a).\end{gathered}\tag{79}$$

Indeed $C_i:x=r_i+at$ meets $A_j:x=r_j-at$ for $t>0$ exactly when $i<j$. There are six vertices, six finite edges and nine exterior rays, with every open edge of multiplicity one. The unique bounded face has boundary

$$Y_2\longrightarrow P_{12}\longrightarrow P_{13}\longrightarrow P_{23}\longrightarrow Y_2.$$

The two other finite edges connect Y_1 to P_{12} and Y_3 to P_{23}. The finite core is connected with one cycle; there are no further crossings, so exactly one face is bounded. Its Euclidean area in the (x,t) plane is $p^2/(2a)$. This is not its area for the form ω.

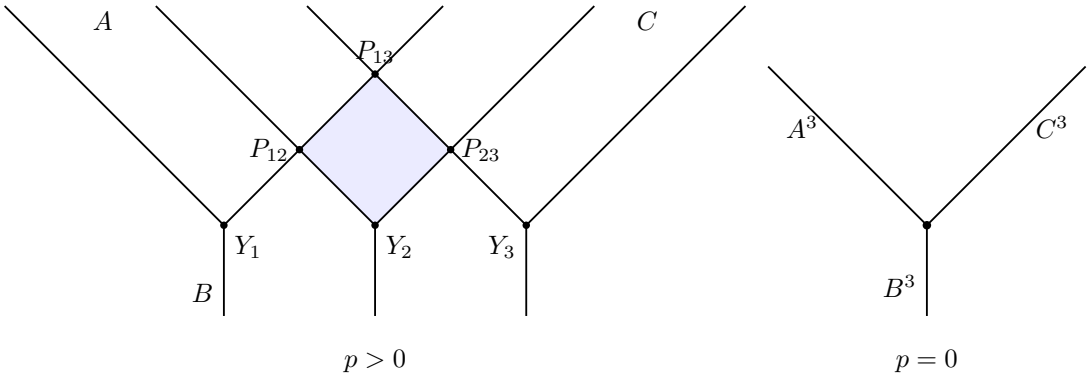


FIGURE 1. The support of (77) for $p > 0$ (left) and $p = 0$ (right). The left panel uses vertical coordinate at/p and horizontal coordinate x/p. Its finite core has six vertices and one bounded diamond. At $p = 0$ the entire finite core collapses to one Y with edge multiplicity three. The four-valent crossings have vertex objects $A \oplus C$.

The vertex object at Y_i is B on the ith coordinate. It has incoming factor B and outgoing filtration $0 \subset A \subset B$, with quotient C. At P_{ij} the object is $A_j \oplus C_i$ with zero arrow, since $i \neq j$. Both filtrations have factors A_j, C_i in that order. Thus a crossing does not create a B strand. These labels and filtrations give an explicit instance of Proposition 7.1.

Proposition 8.2. *For all $p, p' \geq 0$,*

$$\sup_{t\in\mathbb{R}} \max_{v=A,C} d_2(h_v(t;p), h_v(t;p')) \leq \pi\sqrt{2}\,|p - p'|. \tag{80}$$

The same bound holds for each outgoing full boundary norm. At $p = 0$ all three boundary norms are h_0, and the support is the single Y with labels B^3, A^3, C^3.

Proof. Since $0 < \partial_x\lambda(x, s) \leq (1 + s^2)^{-1}$, each integral in (77) gives

$$|w_{vi}(t;p) - w_{vi}(t;p')| \leq \pi|r_i(p) - r_i(p')|$$

uniformly in t. The same basis diagonalizes both norms, so summing the squares of the three weight differences proves (80). Multiplication by the same reference scalar is an isometry; taking normalized limits gives the outgoing bound. At $p = 0$, the incoming weights are zero and the outgoing weights are $A_{\epsilon_v a}(t)$. Normalization removes the latter exactly. This proves the last assertion and displays the collapse explicitly. □

The estimate proves continuity of the full norm paths through this particular topology change. It does not establish uniform continuity for arbitrary boundary triples. Although this example has a bounded face, its norms are a direct sum of scalar norms; bounded faces do not require irreducible boundary data.

8.3. **A triple with no common orthogonal splitting.** Fix $u > 0$. Give the following three bases the weights $(-2u, -u, 0)$:

$$\begin{aligned} b^{(1)} &= (e_1, e_2, e_3), \\ b^{(2)} &= (e_3, e_2, e_1), \\ b^{(3)} &= (e_1 + e_2 + e_3,\ e_1 + 2e_2 + 3e_3,\ e_3). \end{aligned} \tag{81}$$

Their basis matrices are invertible over $\mathbb{C}$. Denote the resulting norms by h_1, h_2, h_3 and the standard norm by h_0. They are valid full boundary data (h_B, h_A^+, h_C^+).

We verify that these norms have neither a common apartment nor a nontrivial common orthogonal direct-sum decomposition. Each is at most h_0. In its residue space the vectors on which $h_i < h_0$ span the plane generated by the first two vectors of $b^{(i)}$. These planes are

$$P_1 = \langle e_1, e_2 \rangle, \quad P_2 = \langle e_2, e_3 \rangle, \quad P_3 = \{x_1 - 2x_2 + x_3 = 0\}.$$

Their common intersection is zero. Rescale any nonzero vector to have h_0-norm one, using the full real value group. Its nonzero residue misses one of the planes, so some h_i has value one on it. Hence $\max_i h_i = h_0$ pointwise.

If the three norms had a common apartment, their maximum would belong to it too. Reducing a common h_0-orthonormal basis would split all three complete residue flags. The first two flags are opposite, with canonical intersection lines $\mathbb{C}e_1, \mathbb{C}e_2, \mathbb{C}e_3$. A basis splitting both must consist of these lines. The first line of the third flag is $\mathbb{C}(e_1 + e_2 + e_3)$, a contradiction.

More generally, a common orthogonal decomposition would be orthogonal for h_0 and would reduce to a decomposition of $\mathbb{C}^3$ splitting the three flags. Its projection preserves the first two opposite flags, so it is a diagonal idempotent in the coordinate basis. Preserving the third flag's first line forces all three diagonal entries to agree. The projection is therefore zero or identity, and one residue summand, hence one original summand, is zero. This excludes a nontrivial decomposition. Theorem 1.2 applies to these triples without reducing them to scalar paths.

8.4. **Full norm information beyond pairwise spectra.** Here is a valued family making that distinction numerical. Set $u = 1$, take $0 < s < 1$, and put $b = T^s$. Keep the first two bases in (81), and replace the third by the columns of

$$B_3(s) = \begin{pmatrix} 1 & 1 & 0 \\ b & 2 & 0 \\ 1 & 3 & 1 \end{pmatrix}.$$

Its determinant $2 - b$ is a unit. Give all three bases weights $(-2, -1, 0)$. Every pairwise logarithmic relative spectrum is $(2, 0, -2)$, independently of s. One can see this from the exterior minors: after including the norm weights the minimum valuations of the entries and of the two-by-two minors are both -2, while the determinant valuation is zero. Unit extremal minors attain these minima; positive-valuation entries cannot decrease them.

For an explicit three-norm invariant, use the standard volume form and let $L_i = \{v : h_i(v) \le 1\}$. Define

$$D_{111}(h_1, h_2, h_3) = -\min\{\nu \det(v_1, v_2, v_3) : v_i \in L_i\}.$$

Each lattice has the finite orthonormal generators $T^{-2}b_1^{(i)}, T^{-1}b_2^{(i)}, b_3^{(i)}$. Multilinearity and the ultrametric inequality reduce the minimum to the finitely many choices of one generator from each lattice. Choosing the first generator in each gives valuation $-6 + s$, because the unscaled determinant is $\pm b$. Every other choice has weight sum at least -5 and an integral determinant. Since $s < 1$, the minimum is $-6 + s$, and therefore

$$D_{111} = 6 - s. \tag{82}$$

Thus the pairwise spectra discard information retained by the full norm triple. The general theorem applies to every member of this valued family. We do not identify its support with the diamond above, or with the dual of a numerical hive built from determinant invariants.

8.5. **Reproducibility.** The accompanying source includes the script `check_examples.py`. It enumerates the nonzero exterior minors of the valued family and verifies their affine valuation inequalities on the whole interval $0 < s < 1$. It also computes the triple determinant minimum and the rational coordinates, incidence relations and area of the normalized diamond. Figure 1 is drawn directly from those coordinates in the accompanying TikZ source. The analytic convergence and continuity arguments are the proofs above and do not depend on quadrature.

9. The matching condition and further questions

9.1. **An edgewise switching example.** The following calculation explains the distinction between the uniqueness proved here and the edgewise conditions printed in [HKKP26, Definition 7.8]. It concerns those displayed conditions; it is not asserted to define a spectral-network object in the sense of [HKS24].

Fix $p > 0$, put $r = (-p, 0, p)$ and $\tau = 3$, and let $I = (-2, -1)$, $J = (1, 2)$. Let σ exchange indices $1, 3$ and fix 2. For $L = I$ or J and $t \le \tau$, define

$$w_i^L(t) = \int_{-\infty}^{t} \left[\mathbf{1}_{s\notin L}\lambda(r_i, s) + \mathbf{1}_{s\in L}\lambda(r_{\sigma(i)}, s)\right] \mathrm{d}s. \tag{83}$$

Use these weights at both vertices in the standard basis. The identity arrow is an isometry and the norms tend to the same standard incoming norm.

The function $\lambda(p, s)$ is even in s. Its integrals over I and J agree. Since the first and third positions are opposites and the second is zero, the two sets of weights in (83) coincide for $2 \le t \le 3$. Write k_i for their common value at $t = \tau$. Continue both paths by

$$w_{vi}(t) = k_i + \int_{\tau}^{t} \lambda(r_i + \epsilon_v a(s - \tau), s)\, \mathrm{d}s, \qquad t \ge \tau.$$

The coordinate gaps are positive for $t > \tau$, and the outgoing derivatives agree with the incoming derivatives at τ. Both normalized outgoing limits exist, by the same integrable difference estimate as in (76). Since the paths coincide on this whole outgoing tail, those full limits are equal.

The two paths are nevertheless different. On $-2 < t < -1$,

$$w_1^I(t) - w_1^J(t) = 2\int_{-2}^{t} \lambda(p, s)\, \mathrm{d}s > 0.$$

Evaluation on e_1 detects their difference as actual norms. At every regular incoming time the unordered positions of either path are $\{-p, 0, p\}$, with stable B labels. At each switch the coordinate subspaces carrying those positions are exchanged. After τ their supports are the same three Ys and their crossings, translated in time from (79). This finite support is connected and balanced. Subdivide at the switches and crossings; all open edges have stable labels and the required straight directions. At a switch both adjacent abstract labels are B, and numerical charge conservation holds.

At a switch, however, the one-sided coordinate velocity vectors in the fixed apartment differ. In the u coordinate, if they are v^-, v^+ at a switch u_0, then the Euclidean apartment formula gives

$$d_2\big(y_v(u_0), \mathrm{mid}(y_v(u_0 - h), y_v(u_0 + h))\big) = \tfrac{h}{2}\left\|v^+ - v^-\right\|_2 + O(h^2).$$

The coefficient is nonzero. Thus both paths fail (10). Equality of support, phases, abstract edge labels and full ends has not identified the embedded velocity subspaces across the exceptional times.

The printed definition describes an edge label as a graded object of a velocity filtration. If that phrase is intended to include additional transport of its embedding, that transport is precisely what must be specified to exclude this example. We have made one sufficient intrinsic condition explicit and proved the correspondence in the resulting finite regular class. The calculation above does not disprove an interpretation incorporating stronger categorical gluing.

9.2. **Comparison with categorical networks.** Proposition 7.1 goes from matched norm paths to local residue objects, two increasing-phase filtrations and their edge quotients. The reverse implication is a separate question: which globally framed network data determine opposite first-order norm germs? In particular, the local existence of two filtrations of an abstract representation does not by itself supply the embedded opposition used in (65).

A comparison with the global Fukaya-category construction would also have to identify the relevant Novikov-field object and its gluing data. The uniqueness theorem of [HKS24, Theorem 4.15], in its constant-coefficient, product-type setting, gives support equality and S-equivalence of reductions for networks representing the same object after Novikov extension. It does not supply this comparison or assert equality of full norm paths. The present variational uniqueness uses the specified boundary triple and the midpoint condition directly.

9.3. **Dependence on the boundary data.** Theorem 1.2 includes every full split boundary triple, and hence includes degenerate inputs. General continuity of the assignment remains open here. A natural topology on triples is the maximum of the three unnormalized d_2 distances. For paths, one may use uniform vertexwise d_2 distance after applying the same prescribed phase scalars on each end. The estimate in Proposition 8.2 establishes this continuity for one explicit family through a bounded-face collapse. A general result would have to control comparisons between variational problems with different full reference norms, rather than only competitors in one fixed-end problem.

Two other extensions are distinct from that continuity problem. First, the theorem fixes the symmetric A_2 charges and radial area form; its finite-support proof uses the three resulting monotone directions. Second, uniqueness has been proved among finite regular paths with matching. Extending it to a larger intrinsically defined class would require a regularity theorem for that class or a suitable direct comparison, even though the minimum itself already has a finite algebraic presentation.

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